MIL-M-38510/175B
30 April 1984
SUPERSEDING
MIL-M-38510/175A
24 November 1980

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, CMOS, POSITIVE LOGIC, FLIP-FLOPS AND MONOSTABLE MULTIVIBRATOR, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, CMOS logic microcircuits. Two product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	4-bit "D" type register
02	Gated J-K master-slave flip-flop (noninverting J-K)
03	Gated J-K master-slave flip-flop (inverting J-K)
04	Dual monostable multivibrator
05	Hex "D" type flip-flop

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510, appendix C)
A	F-1 (14-lead, 1/4" x 1/4"), flat package
C	D-1 (14-lead, 1/4" x 3/4"), dual-in-line package
D	F-2 (14-lead, 1/4" x 3/8"), flat package
E	D-2 (16-lead, 1/4" x 7/8"), dual-in-line package
F	F-5 (16-lead, 1/4" x 3/8"), flat package
X	F-1 (14-lead, 1/4" x 1/4"), flat package, except A dimension = 0.1 (2.54 mm max.)
Y	F-2 (14-lead, 1/4" x 3/8"), flat package, except A dimension = 0.1" (2.54 mm max.)
Z	F-5 (16-lead, 1/4" x 3/8"), flat package, except A dimension = 0.1" (2.54 mm max.)

NOTES:
1. As an exception to 3.5.6.2.3 of MIL-M-38510, for case outlines X, Y, and Z only, the leads of bottom brazed ceramic packages (i.e., configuration 2 of case outlines F-1, F-2, or F-5) may have electroless nickel undercoating which shall be 50 to 200 microinches (1.27 to 5.08 µm) thick provided the lead finish is hot solder dip (i.e., finish letter A) and provided that, after any lead forming, an additional hot solder dip coating is applied which shall extend from the outer tip of the lead to no more than 0.015 inch (0.38 mm) from the package edge.
2. For bottom or side braded packages, case outlines X, Y, and Z only, the S_1 dimension may go to .000 inch (.00 mm) mimimum.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: George C, Marshall Space Flight Center, National Aeronautics and Space Administration, ATTN: EG02, Marshall Space Flight Center, Alabama 35812 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.3 Absolute maximum ratings.

Supply voltage (V_{DD}-V_{SS}) - - -	-0.5 V to +18.0 V
Input current (each input) - - - - -	±10 mA
Input voltage range - - - - - - - - -	(V_{SS} - 0.5) ≤ V_I ≤ (V_{DD} + 0.5)
Storage temperature range- - - - - -	-65°C to +175°C
Maximum power dissipation, P_D- - - -	200 mW
Lead temperature (soldering, 10 seconds) - - - - - - - - - -	300°C
Thermal resistance, junction to case (θ_{JC})- - - - - - - - - - - - -	(See MIL-M-38510 Appendix C)
Junction temperature (T_J)- - - - - -	175°C

1.4 Recommended operating conditions.

Supply voltage (V_{DD}-V_{SS}) - - - - - - - - -	4.5 Vdc to 15.0 Vdc
Input low voltage range (V_{IL})- - - - - - -	0 to 1.5 Vdc @ V_{DD} = 5 Vdc, V_{OL} = 10% V_{DD}, V_{OH} = 90% V_{DD}; 0 to 2.0 Vdc @ V_{DD} = 10 Vdc; 0 to 4.0 Vdc @ V_{DD} = 15 Vdc.
Input high voltage range (V_{IH}) - - - - - -	3.5 to 5.0 Vdc @ V_{DD} = 5 Vdc, V_{OL} = 10% V_{DD}, V_{OH} = 90% V_{DD}; 8.0 to 10.0 Vdc @ V_{DD} = 10 Vdc; 11.0 to 15.0 Vdc @ V_{DD} = 15 Vdc.
Ambient operating temperature range (T_C) -	-55°C to +125°C

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein. Although eutectic die bonding is preferred, epoxy die bonding may be performed. However, the resin used shall be Dupont 5504 Conductive Silver Paste, or equivalent which is cured at 200°C ±10°C for a minimum of 2 hours. The use of equivalent epoxies or cure cycles shall be approved by the qualifying activity. Equivalency shall be demonstrated in data submitted to the qualifying activity for verification.

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3.2.1 Terminal connections. The terminal connections shall be as specified on figure 1.

3.2.2 Logic diagrams. The logic diagrams shall be as specified on figure 2.

3.2.2 Truth tables. The truth tables shall be as specified on figure 3.

3.2.3 Schematic circuits. The schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained and available upon request.

3.2.4 Case outlines. The case outlines shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 and 6.4 herein.

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended ambient operating temperature range, unless otherwise specified.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III. V_{IL}, V_{IH}, and subgroups 7 and 8 testing requires only a summary of attributes data.

3.6 Marking. Marking shall be in accordance with MIL-M-38510.

3.6.1. Total dose radiation hardness identifier. Total dose radiation hardness identifier shall be in accordance with MIL-M-38510 and 4.5.5 herein.

3.6.2 Serialization. All class S devices shall be serialized in accordance with MIL-M-38510.

3.6.3 Correctness of indexing and markings. All devices shall be subjected to the final electrical tests specified in table II after part number marking to verify that they are correctly indexed and identified by part number. Optionally, an approved electrical test may be devised especially for this requirement.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 38 (see MIL-M-38510, appendix E).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Delete the sequence specified in 3.1.9 through 3.1.13 of method 5004 and substitute lines 1 through 7 of table II herein.

b. Burn-in (method 1015 of MIL-STD-883).

(1) Static tests (test condition A) using circuit shown on figure 4, or equivalent. Ambient temperature (T_A) shall be 125°C minimum. Test duration for each static test shall be 24 hours minimum for class S devices and in accordance with table I of method 1015 for class B devices.

(2) Dynamic test (test condition D) using circuit shown on figure 5, or equivalent. Ambient temperature (T_A) shall be 125°C minimum. Test duration shall be in accordance with table I of method 1015.

c. Interim and final electrical parameters shall be as specified in table II herein.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions V_{SS} = 0 V, -55°C ≤ T_A = ≤ 125°C Unless otherwise specified	Device type	Limits		Unit
				Min	Max	
Positive clamping input to V_{DD}	$V_{IC(pos)}$	T_A = 25°C, V_{DD} = GND, V_{SS} = open, Output = open, I_{IN} = 1 mA	All	---	1.5	Vdc
Negative clamping input to V_{SS}	$V_{IC(neg)}$	T_A = 25°C, V_{DD} = open, V_{SS} = GND, Output = open, I_{IN} = -1 mA	All	---	-6	Vdc
Quiescent supply current	I_{SS} or I_{DD}	V_{DD} = 18 Vdc, V_{IN} = V_{SS} or V_{DD} All input combinations	All	---	-2.5	A
High-level output voltage	V_{OH1}	V_{DD} = 15 Vdc, see table III $\|I_O\| \leq 1$ µA	All	14.95	---	Vdc
Low-level output voltage	V_{OL1}	V_{DD} = 15 Vdc, see table III $\|I_O\| \leq 1$ µA	All	---	0.05	Vdc
3-state output current leakage	I_{OC1}	V_{DD} = 18 Vdc, V_O = V_{SS}	01	---	100	nA
	I_{OC2}	V_{DD} = 18 Vdc, V_O = V_{DD}	01	---	-100	nA
Input high voltage	V_{IH1}	V_{DD} = 5 Vdc, see table III $\|I_O\| \leq 1$ µA	All	3.5	---	Vdc
	V_{IH2}	V_{DD} = 10 Vdc, see table III $\|I_O\| \leq 1$ µA	All	7.0	---	Vdc
	V_{IH3}	V_{DD} = 15 Vdc, see table III $\|I_O\| \leq 1$ µA	All	11.0	---	Vdc
Input low voltage	V_{IL1}	V_{DD} = 5 Vdc, see table III $\|I_O\| \leq 1$ µA	All	---	1.5	Vdc
	V_{IL2}	V_{DD} = 10 Vdc, see table III $\|I_O\|$ 1 µA	All	---	3.0	Vdc
	V_{IL3}	V_{DD} = 15 Vdc, see table III $\|I_O\| \leq 1$ µA	All	---	4.0	Vdc
Output high (source) current	I_{OH1}	V_{DD} = 5 Vdc, V_{IN} = V_{DD}, V_{OH} = 4.6 Vdc	All	-0.36	---	mAdc
	I_{OH2}	V_{DD} = 15 Vdc, V_{IN} = V_{DD}, V_{OH} = 13.5 Vdc	All	-2.4	---	mAdc
Output low (sink) current	I_{OL1}	V_{DD} = 5 Vdc, see table III V_{OL} = 0.4 Vdc	All	0.36	---	mAdc
	I_{OL2}	V_{DD} = 15 Vdc, see table III V_{OL} = 1.5 Vdc	All	2.4	---	mAdc

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions V_{SS} = 0 V, $-55°C \leq T_A = \leq 125°C$, Unless otherwise specified	Device type	Limits Min	Limits Max	Unit
Input leakage current 1/	I_{IH}	V_{DD} = 18 Vdc. Measure inputs sequentially, connect all unused inputs to V_{DD} or V_{SS}.	All	---	45	nA
Input leakage current 1/	I_{IL}	V_{DD} = 18 Vdc. Measure input sequentially, connect all unused inputs to V_{DD} or V_{SS}.	All	---	-45	nA
Input capacitance	C_i	V_{DD} = 0 Vdc, f = 1 MHz, T_A = 25°C	02,03, 04 01,05	---	7.5 15	pF
Input test voltage	V_{ZAP}	C_1 = 100 pF, R_2 = 1.5 kΩ (see 4.5.3)	All	400	---	V
Propagation delay times (clock to output) High to low Low to high	t_{PHL1} t_{PLH1}	V_{DD} = 5 Vdc, C_L = 50 pF, R_L = 200 kΩ (see figure 6)	01 02,03, 04 05	30 25 15	840 700 420	ns
Propagation delay times (reset, set, or clear to output) High to low Low to high	t_{PHL2} t_{PLH2} 2/	V_{DD} = 5 Vdc, C_L = 50 pF, R_L = 200 kΩ (see figure 6)	01 02,03, 05 04	23 15 23	645 420 630	ns
Transition times High to low Low to high	t_{THL} t_{TLH}	V_{DD} = 5 Vdc, C_L = 50 pF, R_L = 200 kΩ (see figure 6) See table III for device type 04	All	10	280	ns
3-state delay times Output high-to-high impedance Output low-to-high impedance High-impedance to output high High-impedance to output low	t_{PHZ} t_{PLZ} t_{PZH} t_{PZL}	V_{DD} = 5 Vdc, C_L = 50 pF, R_L = 1 kΩ (see figure 6)	01	15	420	ns
Minimum clock transition times High to low Low to high	$t_{THL(CLK)}$ $t_{TLH(CLK)}$	V_{DD} = 5 Vdc, C_L = 50 pF, R_L = 200 kΩ (see figure 6)	01,02, 03,05	---	15	µs
Minimum data setup times High to low Low to high	t_{SHL} t_{SLH}	V_{DD} = 5 Vdc, C_L = 50 pF, R_L = 200 kΩ (see figure 6)	01 02,03 05	--- --- ---	280 560 120	ns

1/ Input current at one input node.
2/ Device types 01 and 05 do not have a t_{PLH2} test.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions V_{SS} = 0 V -55°C ≤ T_A = ≤ 125°C Unless otherwise specified	Device type	Limits		Unit
				Min	Max	
Minimum data hold times High to low Low to high	 t_{HHL} t_{HLH}	V_{DD} = 5 Vdc, C_L = 50 pF, R_L = 200 kΩ (see figure 6)	05	---	85	ns
Minimum pulse widths Reset Set Clock Clear	 t_{PR} t_{PS} $t_{p(CLK)}$ $t_{p(CLR)}$	V_{DD} = 5 Vdc, C_L = 50 pF, R_L = 200 kΩ (see figure 6)	01 02,03 02 01,05 02,03 05	--- --- --- --- --- ---	170 280 280 280 200 280	ns
Maximum clock frequency	f_{CLK}	V_{DD} = 5 Vdc, C_L = 50 pF, R_L = 200 kΩ (see figure 6)	01 02,03 05	--- --- ---	1,000 1,400 800	ns
Propagation delay times (clock to output) High to low Low to high	 t_{PHL1} t_{PLH1}	V_{DD} = 10 Vdc, C_L = 50 pF, R_L = 200 kΩ, T_A = 25°C (see figure 6)	01,02,04 03 05	13 10 5	250 200 110	ns
Propagation delay times (reset, set, or clear to output) High to low Low to high	 t_{PHL2} t_{PLH2} 2/	V_{DD} = 10 Vdc, C_L = 50 pF, R_L = 200 kΩ, T_A = 25°C (see figure 6)	01 02,03 04 05	10 8 13 5	200 150 250 110	ns
Transition times High to low Low to high	 t_{THL} t_{TLH}	V_{DD} = 10 Vdc, C_L = 50 pF, R_L = 200 kΩ, T_A = 25°C (see figure 6) See table III for device type 04	01,02,03,04 05	5 5	100 100	ns
3-state delay times Output high-to-high impedance Output low-to-high impedance High-impedance to output high High-impedance to low output	 t_{PHZ} t_{PLZ} t_{PZH} t_{PZL}	V_{DD} = 10 Vdc; C_L = 50 pF, R_L = 1 kΩ, T_A = 25°C (see figure 6)	01	8	150	ns

2/ Device types 01 and 05 do not have a t_{PLH2} test.

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TABLE II. Electrical test requirements.

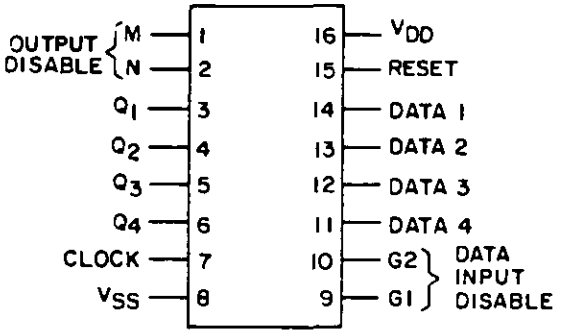
Line no.	MIL-STD-883 test requirements	Class S device 3/				Class B device 3/			
		Ref par.	Table 2/ III sub-groups	Table 1/ IV delta limits	Test circuit figure	Ref par.	Table 2/ III sub-groups	Table 1/ IV delta limits	Test circuit figure
1	Interim electrical parameters method 5004		1				1		
2	Static burn-in I method 1015	4.2b 4.5.2			4				
3	Same as line 1		1	Δ					
4	Static burn-in II method 1015	4.2b 4.5.2			4	4.2b 4.5.2	4/		4
5	Same as line 1		1*	Δ		4.2d	1*		
6	Dynamic burn-in method 1015	4.2b 4.5.2			5				
7	Same as line 1	4.2d	1*	Δ					
8	Final electrical parameters method 5004		1*,2,3, 7,9				1*,2,3, 7, 9		
9	Group A test requirements method 5005	4.4.1	1,2,3,4, 7,8,9, 10,11,12			4.4.1	1,2,3,4, 7,9		
10	Group B end-point electrical parameters method 5005	4.4.2	1,2,3,7, 8,9,10,11	Δ					
11	Group C end-point electrical parameters method 5005					4.4.3	1,2,3	Δ	
12	Additional group C tests method 5005					4.4.3c	10,11		
13	Group D end-point electrical parameters method 5005	4.4.4	1,2,3			4.4.4	1,2,3		

1/ (Δ)indicates delta limit shall be required only on table III subgroup 1, where specified, and delta values shall be computed with reference to the previous interim electrical parameters (line 1).
2/ (*) indicates PDA applies to subgroup 1 (see 4.2.1).
3/ Blank spaces indicate tests are not applicable.
4/ The device manufacturer may at his option either perform delta measurements or within 24 hours after burn-in (or removal of bias) perform the final electrical parameter measurements.

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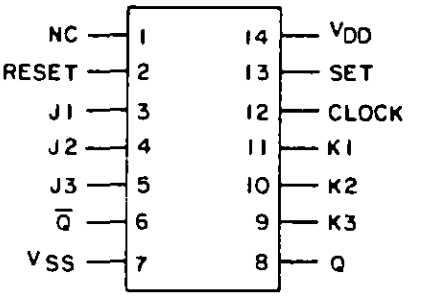
Device type 01

Cases E,F,and Z



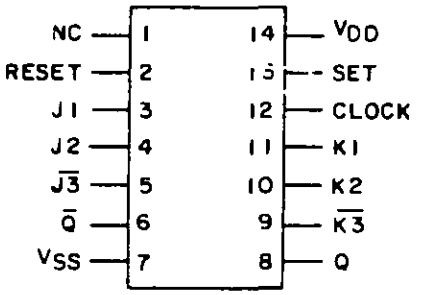
Device type 02

Cases A,C,D,X,and Y



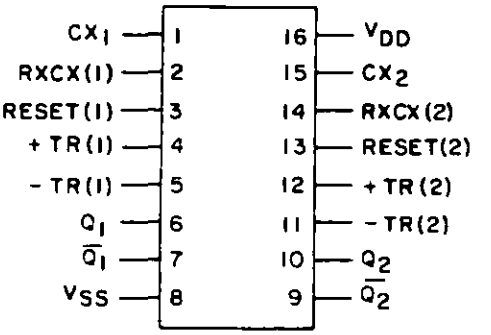
Device type 03

Cases A,C,D,X,and Y



Device type 04

Cases E,F,and Z



Device type 05

Cases E,F,and Z

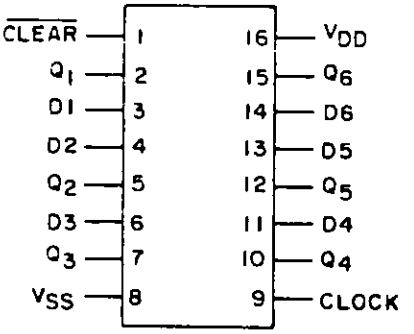
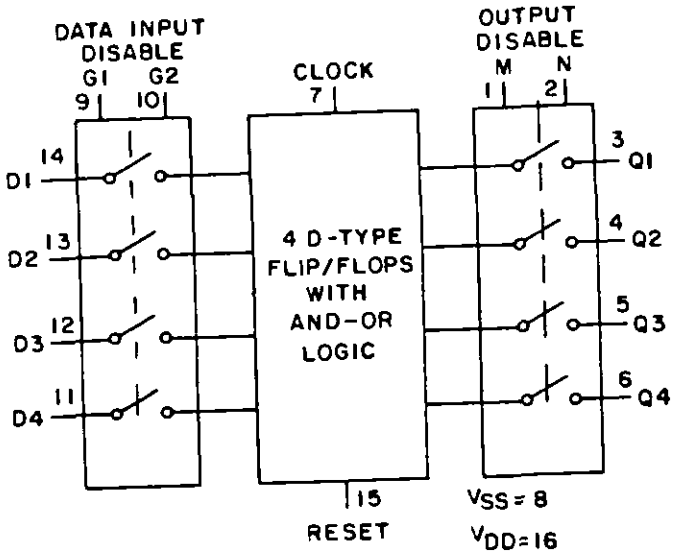


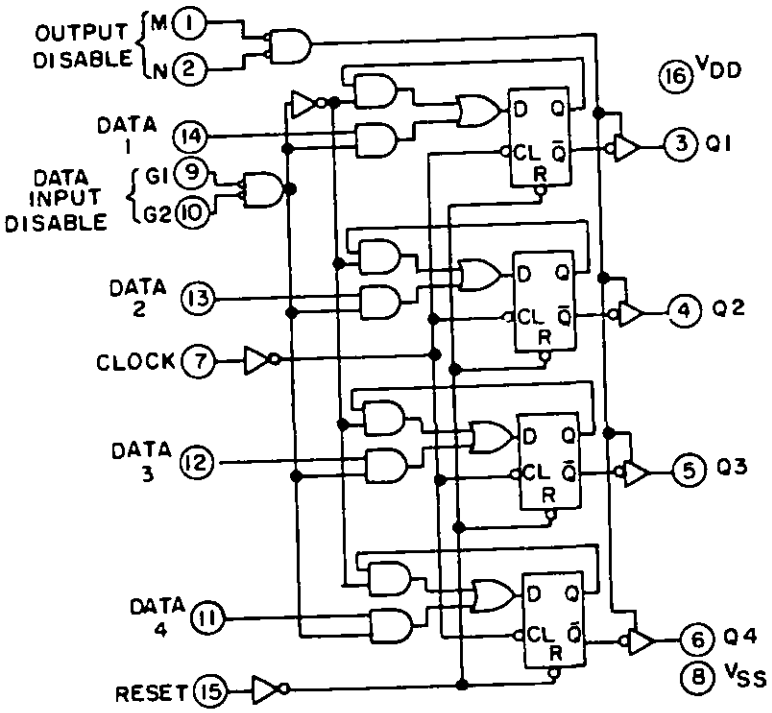
FIGURE 1. Terminal connections.

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Device type 01



FUNCTIONAL DIAGRAM

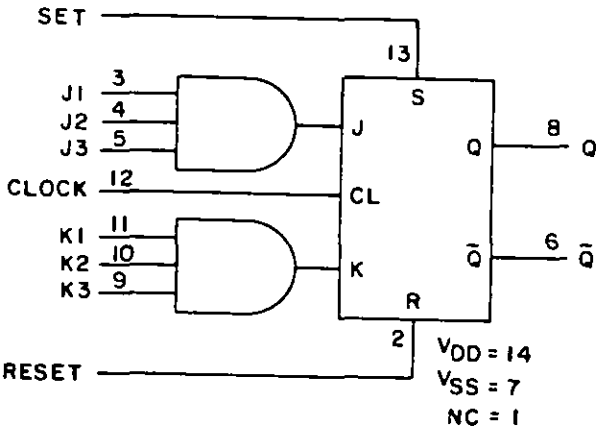


LOGIC DIAGRAM

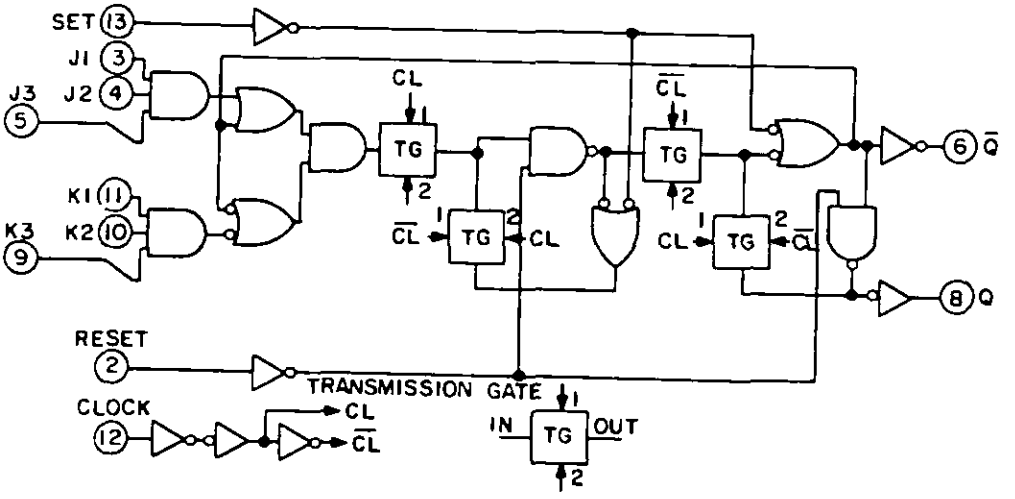
FIGURE 2. Logic diagrams.

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Device type 02



FUNCTIONAL DIAGRAM



LOGIC DIAGRAM

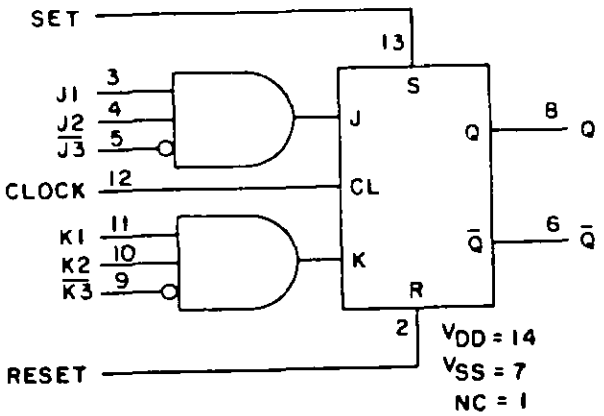
NOTE: Input to output is:

a. A bidirectional low impedance when control input 1 is "low" and control input 2 is "high".
b. An open circuit when control input 1 is "high" and control input 2 is "low".

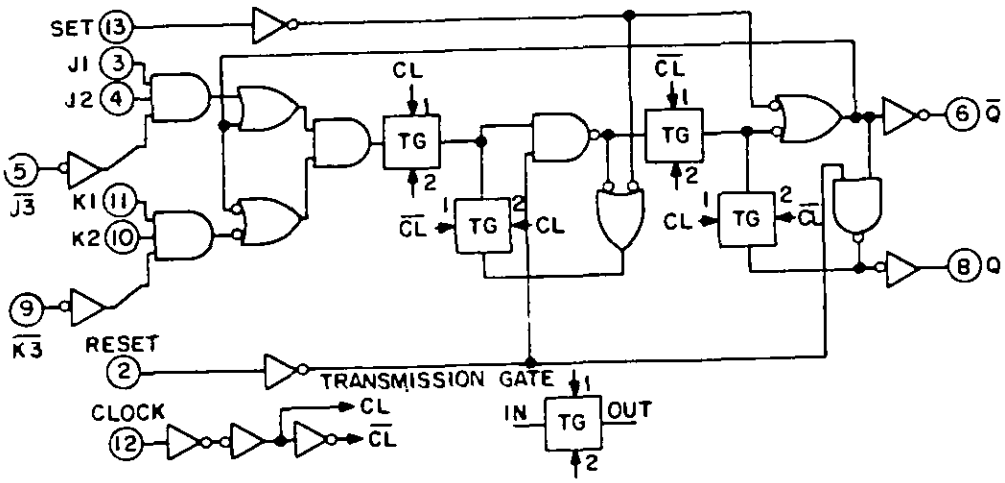
FIGURE 2. Logic diagrams - Continued.

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Device type 03



FUNCTIONAL DIAGRAM



LOGIC DIAGRAM

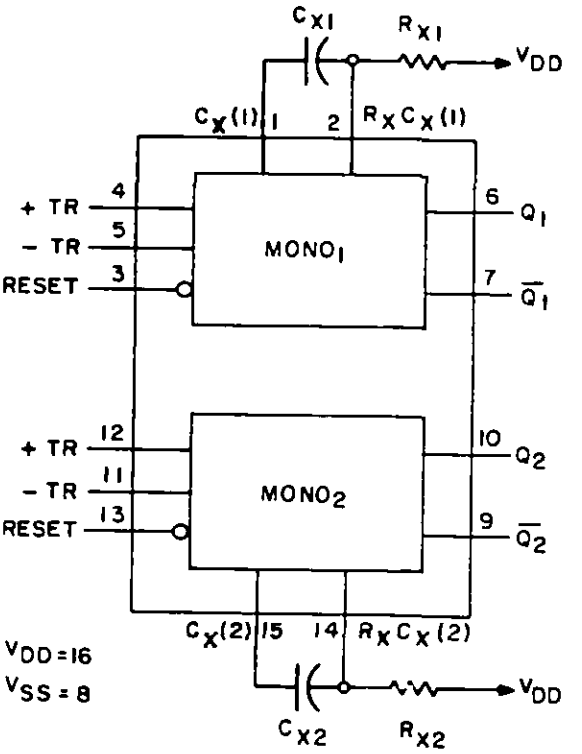
NOTE: Input to output is:

a. A bidirectional low impedance when control input 1 is "low" and control input 2 is "high".
b. An open circuit when control input 1 is "high" and control input 2 is "low".

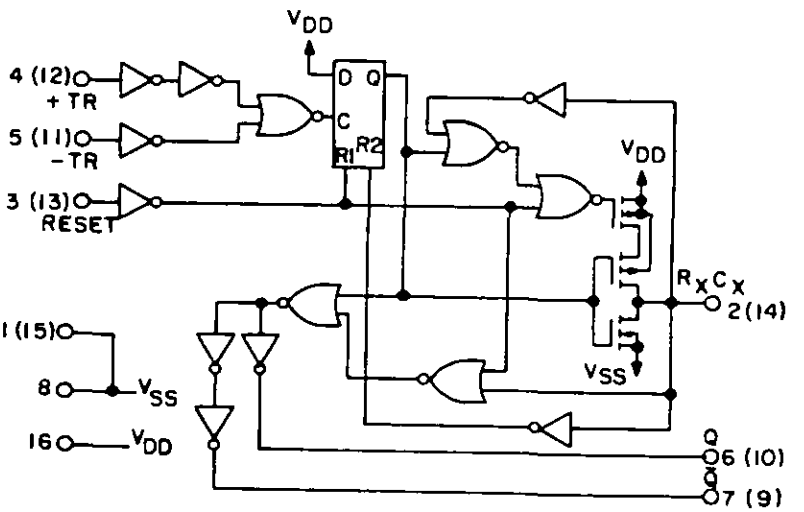
FIGURE 2. Logic diagrams - Continued.

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Device type 04



FUNCTIONAL DIAGRAM



LOGIC DIAGRAM

NOTE: Schematic shown is 1/2 of total package. Terminals 1, 8, and 15 are electrically connected internally.

FIGURE 2. Logic diagrams - Continued.

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Device type 05

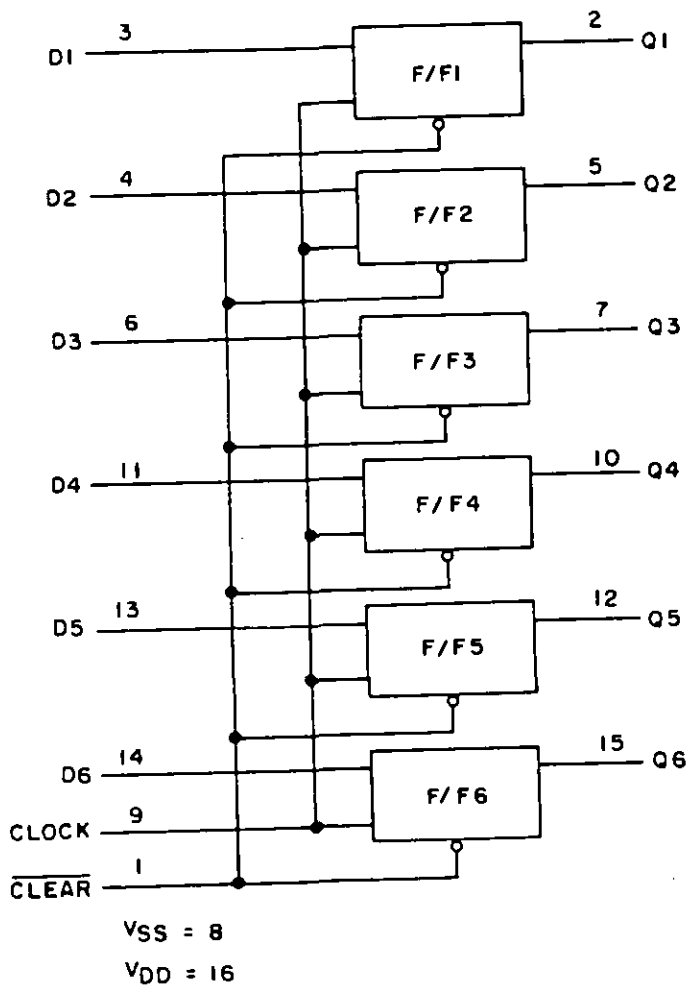


FIGURE 2. Logic diagrams - Continued.

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Device type 01

Truth Table						
Reset	Clock	Data Input Disable G1	Data Input Disable G2	Data D	Next State Output Q	
H	X	X	X	X	L	
L	L	X	X	X	Q	NC
L	↑	H	X	X	Q	NC
L	↑	X	H	X	Q	NC
L	↑	L	L	H	H	
L	↑	L	L	L	L	
L	H	X	X	X	Q	NC
L	↓	X	X	X	Q	NC

When either Output Disable M or N is high, the outputs are disabled (high impedance state), however sequential operation of the flip-flops is not affected.

H = High Level
L = Low Level
X = Don't Care
NC = No Change

Device types 02 and 03

(Synchronous operation S = 0, R = 0)

Truth Table			
Inputs before positive clock transition		Outputs after positive clock transition	
Inputs		Outputs	
J*	K*	Q	$\overline{Q}$
L	L	N/C	N/C
L	H	L	H
H	L	H	L
H	H	Toggles	

*Device type 02

For type 02

$J = J_1 \cdot J_2 \cdot J_3$

$K = K_1 \cdot K_2 \cdot K_3$

For type 03

$J = J_1 \cdot J_2 \cdot \overline{J_3}$

$K = K_1 \cdot K_2 \cdot \overline{K_3}$

Asynchronous Operation

(J and K Don't Care)

Truth Table			
Inputs		Outputs	
S	R	Q	$\overline{Q}$
L	L	N/C	N/C
L	H	L	H
H	L	H	L
H	H	L	L

H = High Level
L = Low Level
NC = No Change

FIGURE 3. Truth tables.

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Device type 04

TRUTH TABLE					
INPUTS			OUTPUTS		TIMING
+ TRIGGER	_ TRIGGER	RESET	Q	$\bar{Q}$	R_X C_X
X	X	L	L	H	X
⌐	H	H	H	L	H
¬	H	H	H	L	H
X	X	H	H	L	L
L	¬	H	H	L	H
L	⌐	H	H	L	H

X=DON'T CARE
H=HIGH LEVEL
L=LOW LEVEL

Device type 05

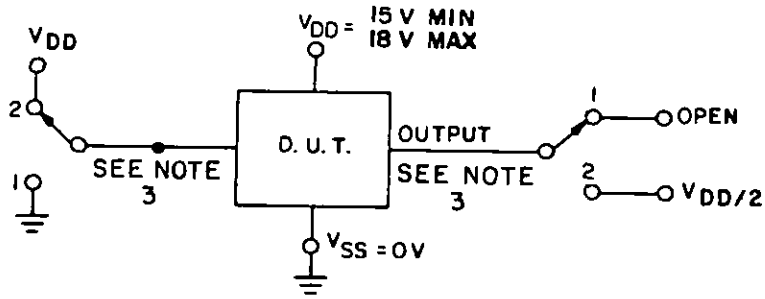
Truth Table for 1 of 6 flip-flops			
Inputs			Output
Clock	Data	$\overline{\text{Clear}}$	Q
⌐	L	H	L
⌐	H	H	H
¬	X	H	N/C
X	X	L	L

H = High Level
L = Low Level

X = Don't Care
N/C = No Change

FIGURE 3. Truth tables - Continued.

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NOTES:
1. For static burn-in I, all inputs are connected to 0 volts, switch position 1.
2. For static burn-in II, all inputs are connected to V_{DD}, switch position 2.
3. Except for V_{DD} and V_{SS}, each terminal shall be connected through a resistor whose value is 2 kΩ to 47 kΩ. The actual measured value of the resistor selected shall not exeed ±20% of its branded value due to use, heat or age.
4. Output may be in switch position 1 or 2.

FIGURE 4. Static burn-in test circuits.

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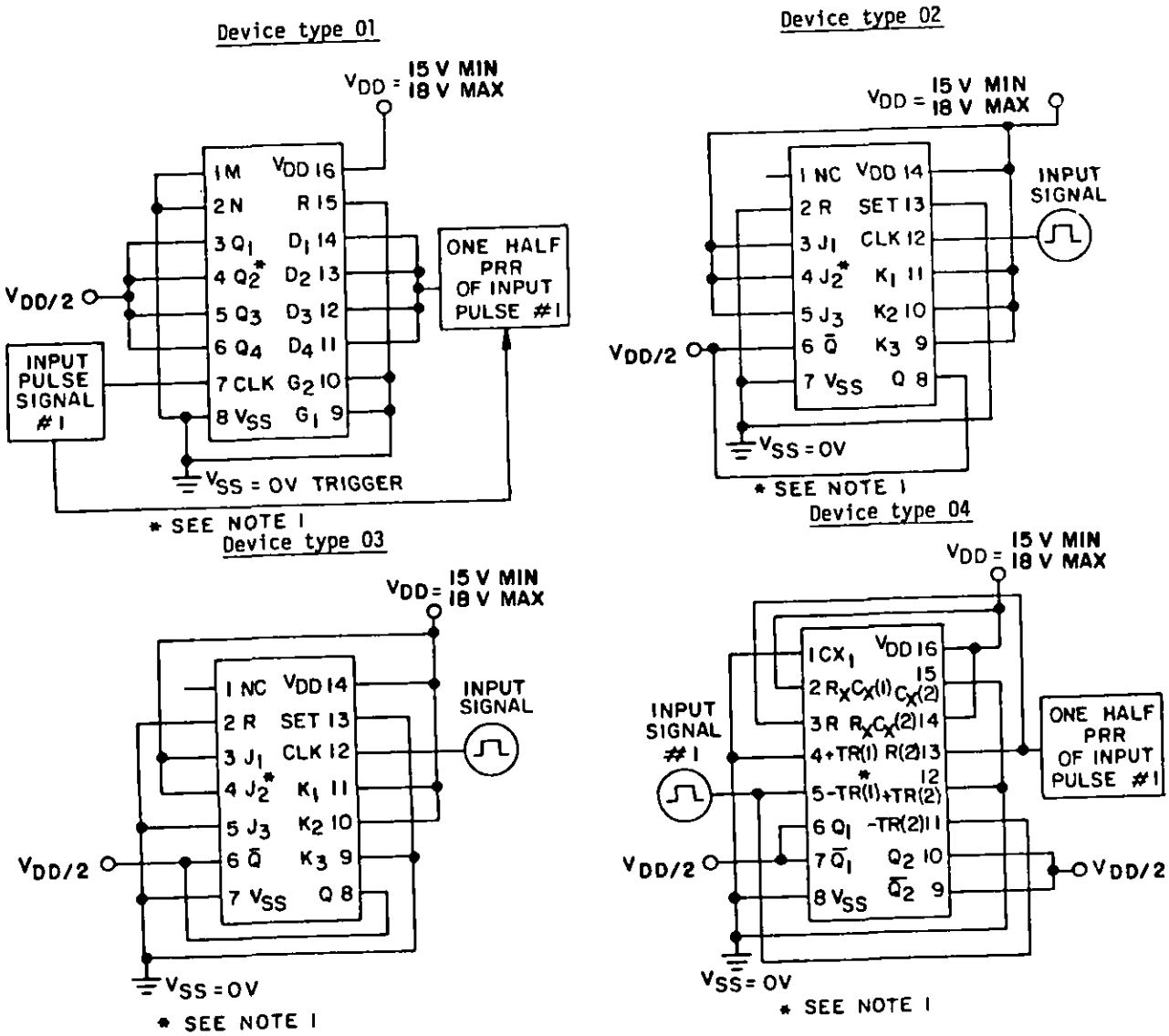
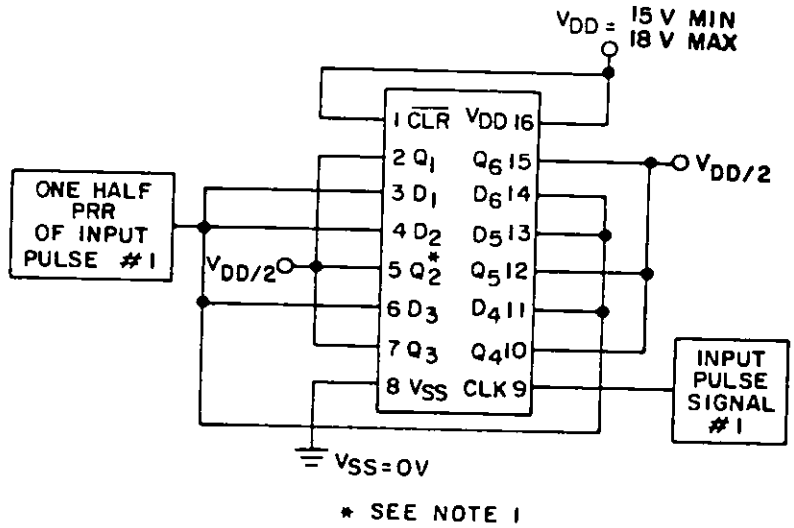


FIGURE 5. Dynamic burn-in and steady state life test circuit.

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Device type 05



NOTES:

1. Except for V_{DD} and V_{SS}, each terminal shall be connected through a resistor whose value is 2 kΩ to 47 kΩ. The actual measured value of the resistor selected shall not exeed ±20% of its branded value due to use, heat or age.
2. Input pulse signal #1 requirements:

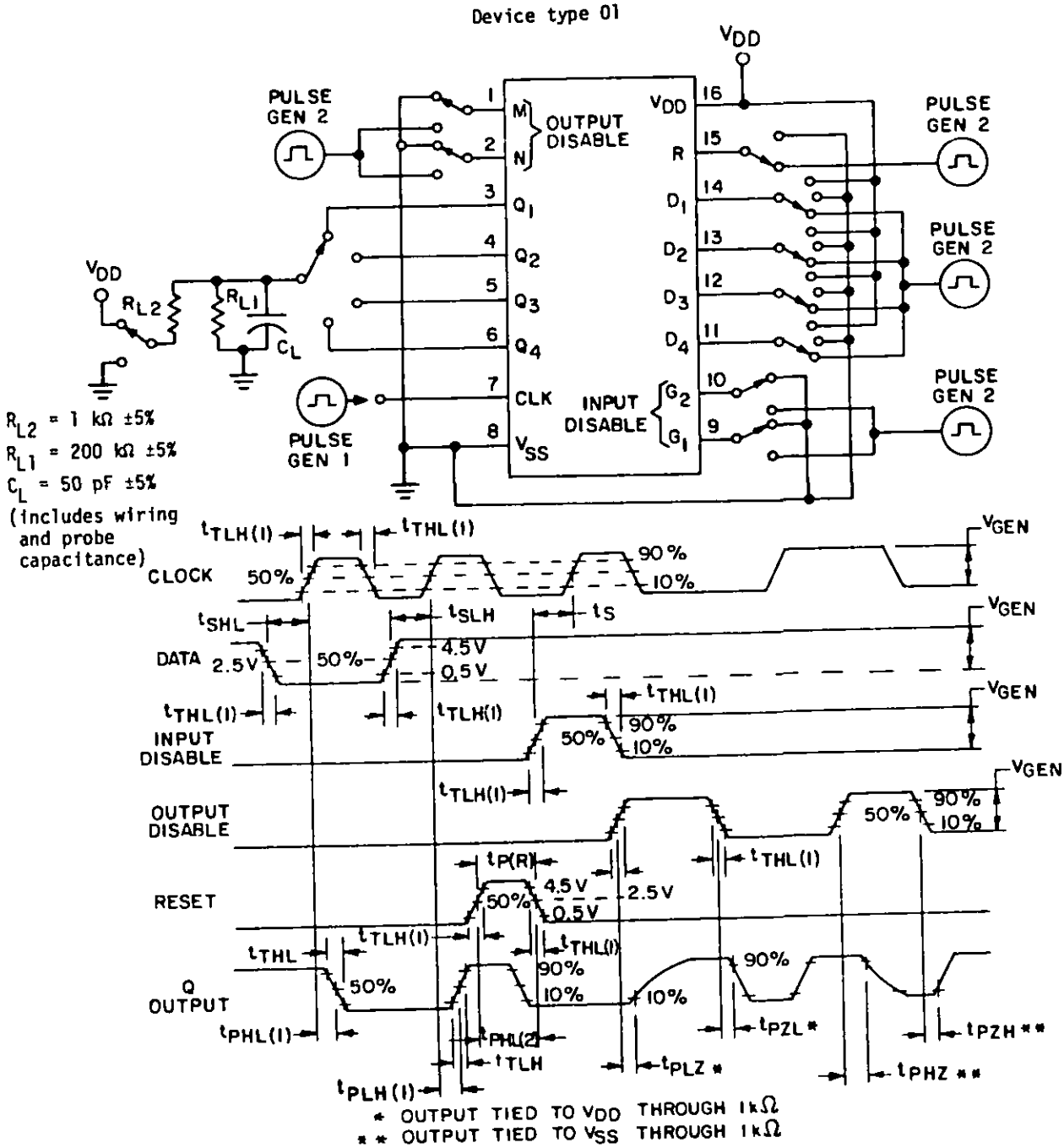
   a. Square wave, 50% duty cycle.
   b. 25 kHz < PRR < MHz.
   c. t_{TLH} and t_{THL} < 1 µs.
   d. Voltage = 0 to 15 V minimum to 18 V maximum peak.

3. Device type 05 - Positive transition of F occurs at center of F/2.

FIGURE 5. Dynamic burn-in and steady state life test circuit - Continued.

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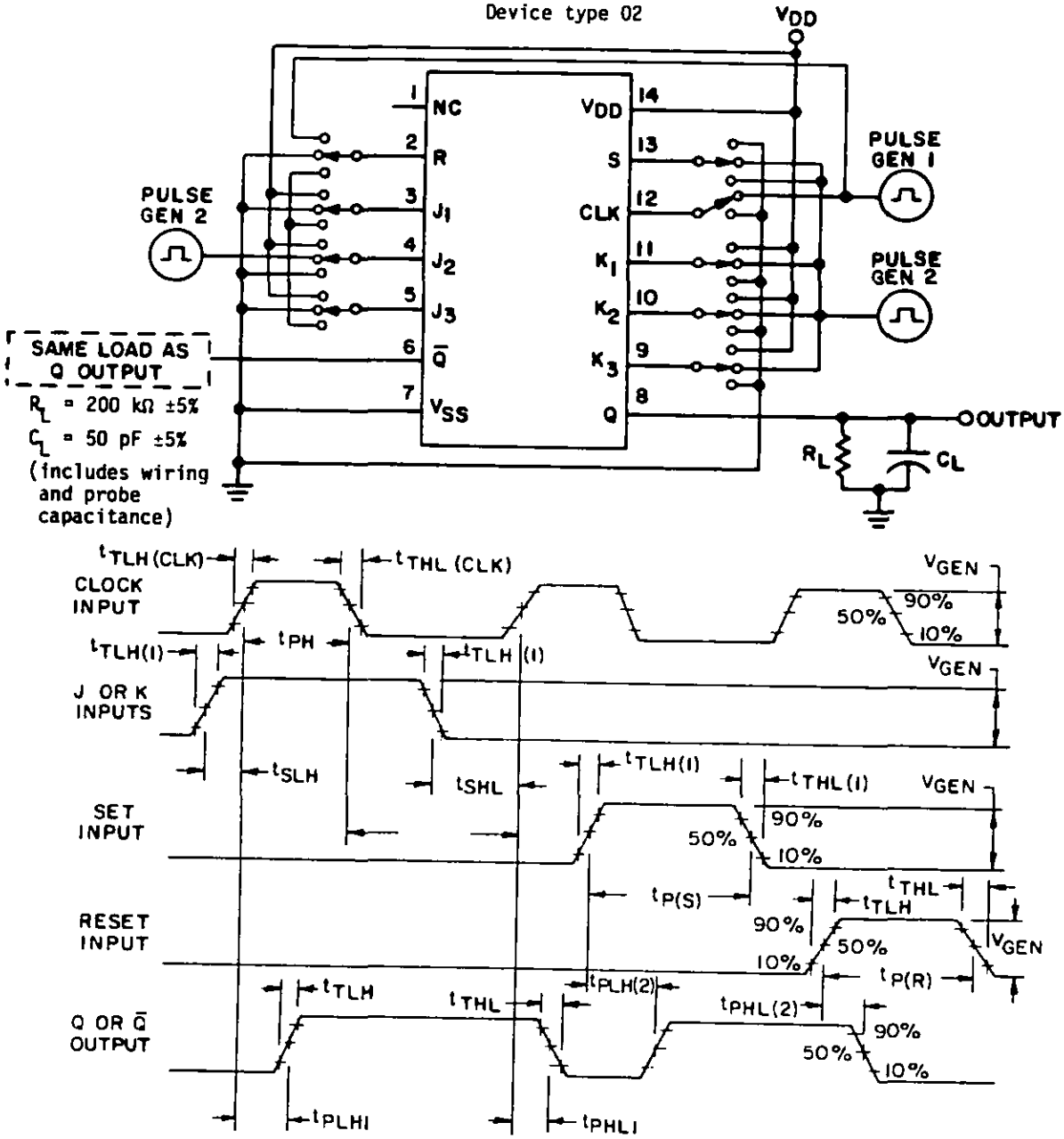
Device type 01



NOTES:
1. Pulse generator number 1 has the following characteristics: $V_{GEN} = V_{DD} \pm 1\%$, $t_{PH} = 1.0 \pm 0.1$ µs, $t_{THL(1)}$ and $t_{TLH(1)} = 20 \pm 2$ ns and PRR = 200 KHz.
2. Pulse generators number 2 have the following characteristics: $V_{GEN} = V_{DD} \pm 1.0\%$, $t_{PH} = 1.0 \pm 0.1$ µs, $t_{THL(1)}$ and $t_{TLH(1)} = 20 \pm 2$ ns and PRR = 100 KHz.
3. Requirements for minimum clock frequency (f_{CL}), $t_{TLH(CL)}$, $t_{THL(CL)}$, t_{SHL}, t_{SLH}, $t_{p(CL)}$, and $t_{p(R)}$ are established by setting the parameter to the limits given in table III and observing proper output state change.
4. Complete terminal conditions are as specified in table III.

FIGURE 6. Switching test circuit and waveforms.

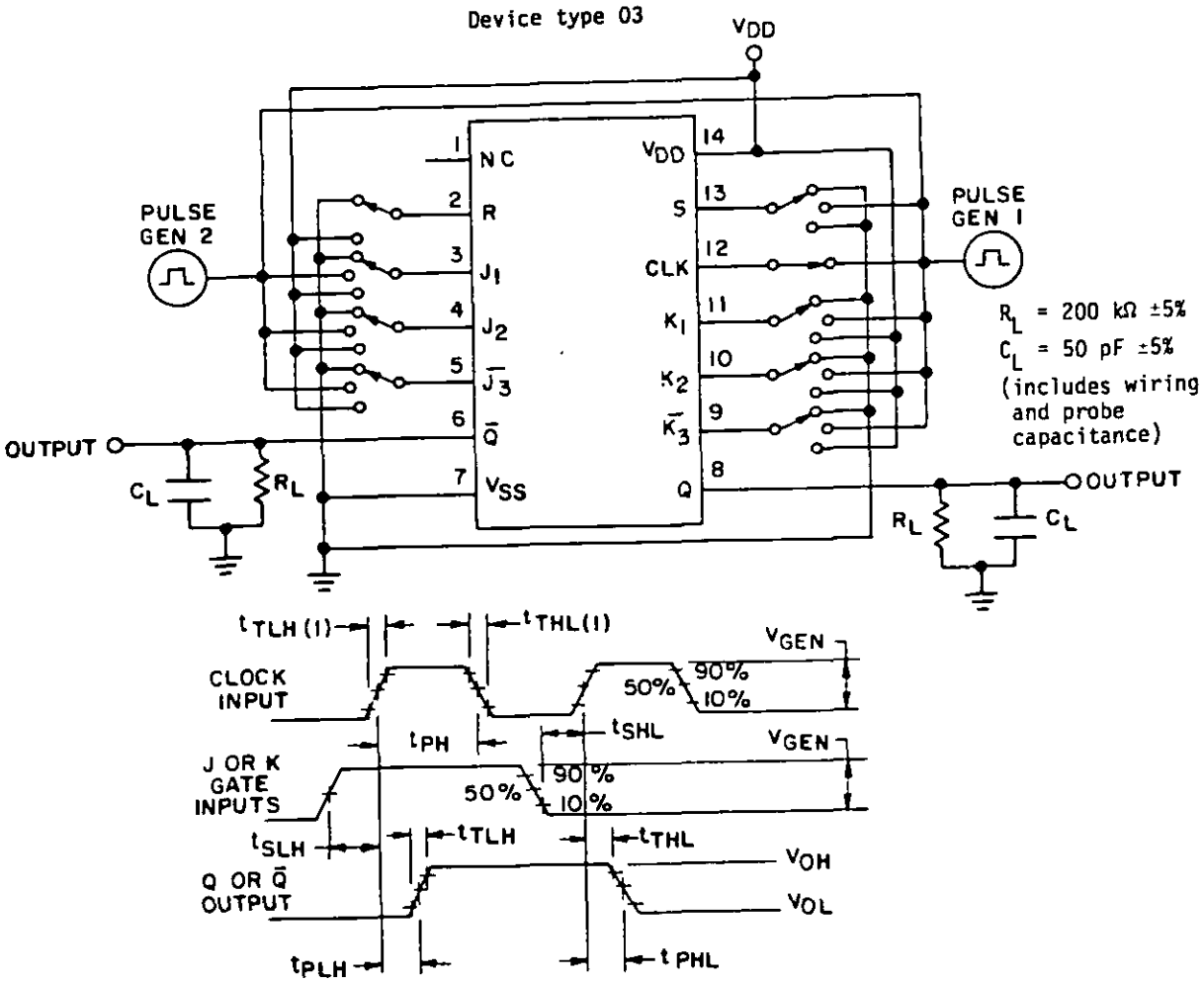
Source: https://assist.dla.mil -- Downloaded: 202640-21T13:44Z
Check the source to verify that this is the current version before use.



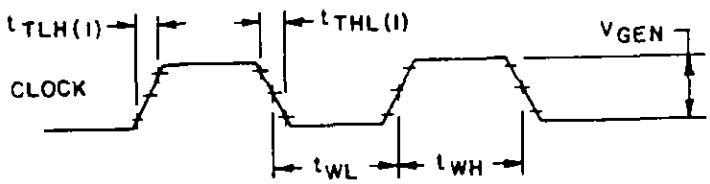
NOTES:
1. Pulse generator number 1 has the following characteristics: $V_{GEN} = V_{DD} \pm 1\%$, $t_{PH} = 1.0 \pm 0.1$ µs, $t_{THL(1)}$ and $t_{TLH(1)} = 20 \pm 2$ ns, PRR = 200 KHz.
2. Pulse generator number 2 has the following characteristics: $V_{GEN} = V_{DD} \pm 1\%$, $t_{PH} = 1.0 \pm 0.1$ µs, $t_{THL(1)}$ and $t_{TLH(1)} = 20 \pm 2$ ns, PRR = 100 KHz.
3. Requirements for minimum clock frequency (f_{CLK}), $t_{TLH(CLK)}$, $t_{THL(CLK)}$, $t_{p(CLK)}$, $t_{p(R)}$, t_{SL}, and t_{SH} are established by setting the parameter to the limits given in table III and observing proper output state change.
4. Complete terminal conditions are as specified in table III.

FIGURE 6. Switching test circuit and waveforms - Continued.

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PROPAGATION DELAY, TRANSITION, AND SETUP-TIME WAVEFORMS.



CLOCK PULSE RISE AND FALL TIME WAVEFORMS.

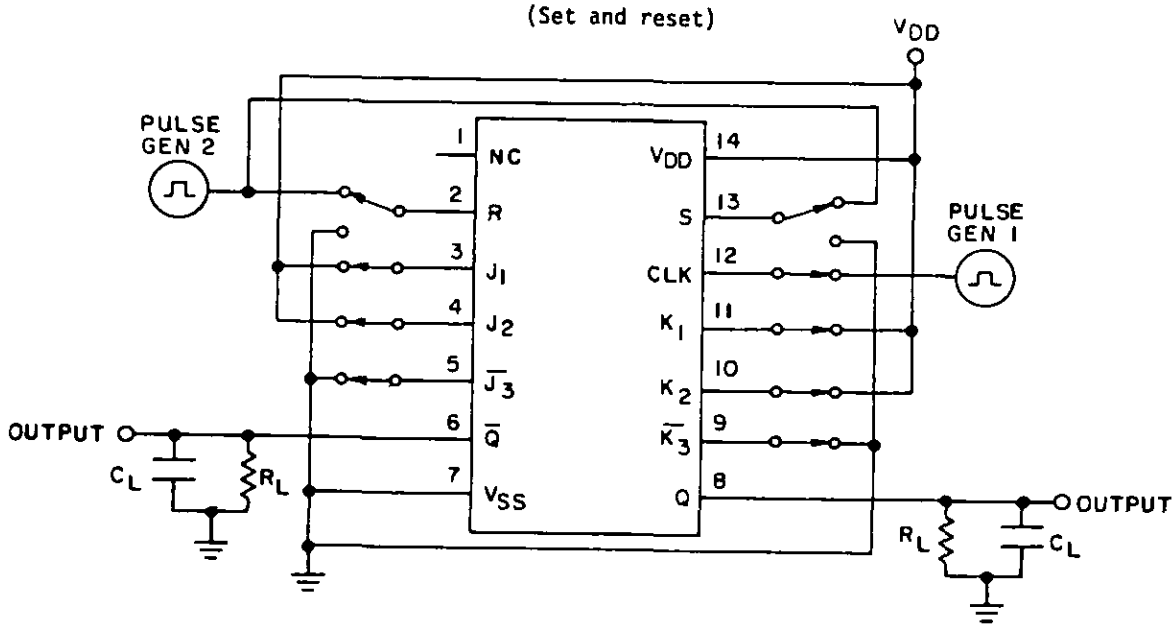
NOTES:
1. Pulse generator number 1 has the following characteristics: $V_{GEN} = V_{DD} \pm 1\%$, $t_{PH} = 1.0 \pm 0.1$ µs, $t_{THL(1)}$ and $t_{TLH(1)} = 20 \pm 2$ ns, PRR = 200 KHz.
2. Pulse generator number 2 have the following characteristics: $V_{GEN} = V_{DD} \pm 1\%$, $t_{PH} = 1.0 \pm 0.1$ µs, $t_{THL(1)}$ and $t_{TLH(1)} = 20 \pm 2$ ns, PRR = 100 KHz.
3. Requirements for minimum clock frequency (f_{CLK}), $t_{TLH(CLK)}$, $t_{THL(CLK)}$, $t_{p(CLK)}$, $t_{p(R)}$, t_{SL}, and t_{SH} are established by setting the parameter to the limits given in table III and observing proper output state change.
4. Complete terminal conditions are as specified in table III.

FIGURE 6. Switching test circuit and waveforms - Continued.

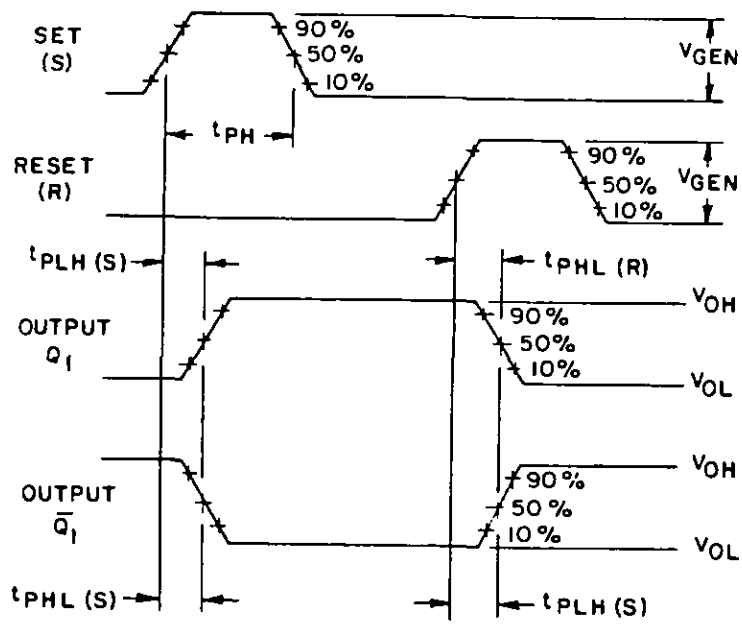
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Device type 03

(Set and reset)



R_L = 200 kΩ ±5%.
C_L = 50 pF ±5% (includes wiring and probe capacitance).



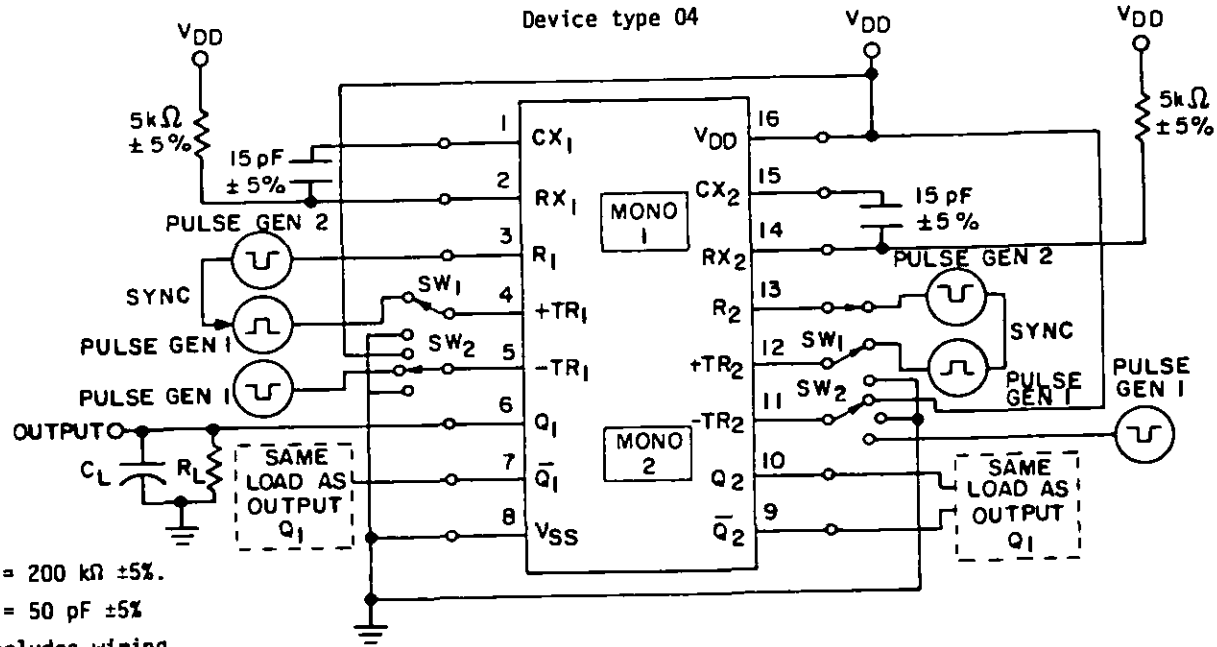
NOTES:

1. Pulse generator number 1 has the following characteristics: $V_{GEN} = V_{DD} \pm 1\%$, t_{PH} = 1.0 ±0.1 µs, $t_{THL(1)}$ and $t_{TLH(1)}$ = 20 ±2 ns and PRR = 200 KHz.
2. Pulse generator number 2 has the following characteristics: $V_{GEN} = V_{DD} \pm 1\%$, t_{PH} = 1.0 ±0.1 µs, $t_{THL(1)}$ and $t_{TLH(1)}$ = 20 ±2 ns and delayed 2.0 µs after pulse generator number 1.
3. Complete terminal conditions are specified in table III.

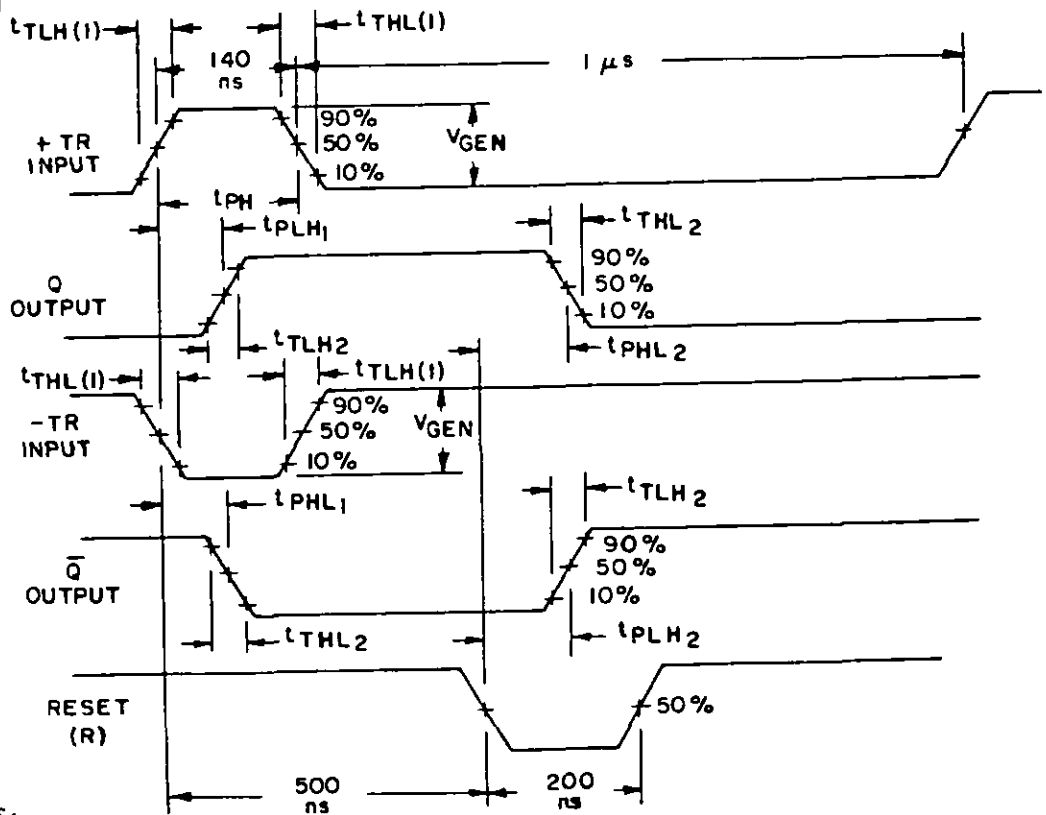
FIGURE 6. Switching test circuit and waveforms - Continued.

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Device type 04



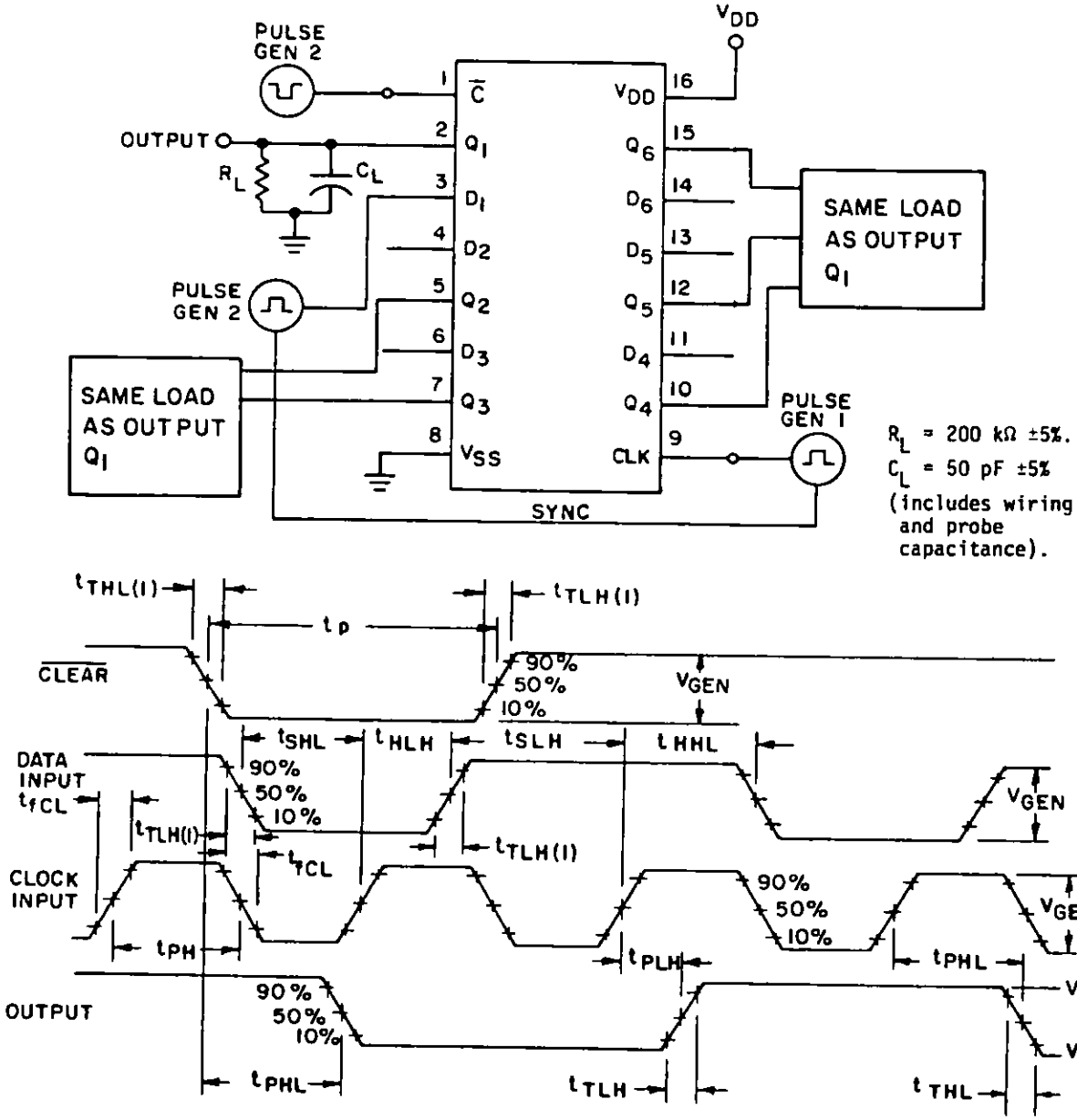
R_L = 200 kΩ ±5%.
C_L = 50 pF ±5%
(includes wiring and probe capacitance).



NOTES:
1. Pulse generator number 1 has the following characteristics: $V_{GEN} = V_{DD} \pm 1.0\%$, t_p as specified in +TR and -TR waveforms, $t_{TLH(1)}$ and $t_{THL(1)}$ = 20 ±2 ns and PRR = 877.193 kHz.
2. Pulse generator number 2 has the following characteristics: $V_{GEN} = V_{DD} \pm 1.0\%$, $t_{TLH(1)}$ and $t_{THL(1)}$ = 20 ±2 ns and PRR = 877.193 kHz.
3. Identical switching measurements are obtained from mono 1 and mono 2.
4. Complete terminal conditions are specified in table III.

FIGURE 6. Switching test circuit and waveforms - Continued.

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NOTES:
1. Pulse generator number 1 has the following characteristics: $V_{GEN} = V_{DD}$ ±1%, t_{PH} = 1.0 ±0.1 µs, $t_{TLH(1)}$ and $t_{THL(1)}$ = 20 ±2 ns and PRR = 200 kHz.
2. Pulse generator number 2 has the following characteristics: $V_{GEN} = V_{DD}$ ±1%, t_{PH} = 1.0 ±0.1 µs, $t_{TLH(1)}$ and $t_{THL(1)}$ = 20 ±2 ns and PRR = 100 kHz.
3. Requirements for minimum clock frequency (f_{CLK}), $t_{TLH(CLK)}$, $t_{THL(CLK)}$, $t_{p(CLK)}$, $t_{p(CLR)}$, t_{SLH}, t_{SHL}, t_{HHL} and t_{HLH} are established by setting the parameter to the limits given in table III and observing proper output state change.
4. Identical measurements are obtained from flip flop 1 thru 6.

FIGURE 6. Switching test circuit and waveforms - Continued.

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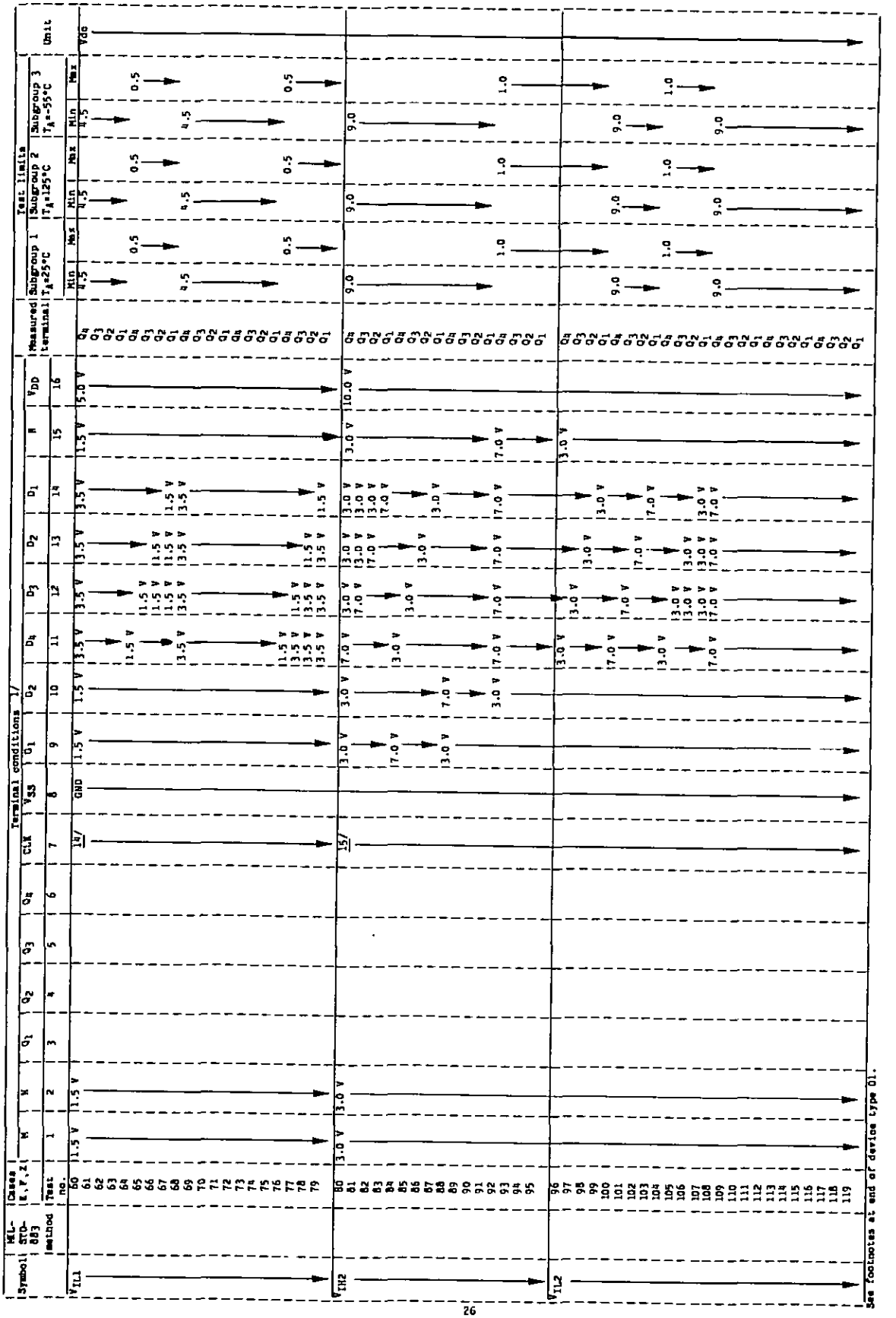
TABLE III. Group A inspection for device type 01.

Symbol	MIL-STD-883 method	Cases E, F, Z	Terminal conditions 1/ M	N	Q_1	Q_2	Q_3	Q_4	CLK	V_{SS}	G_1	G_2	D_4	D_3	D_2	D_1	R	V_{DD}	Measured terminal	Test limits Subgroup 1 T_A=25°C Min	Subgroup 1 T_A=25°C Max	Subgroup 2 T_A=125°C Min	Subgroup 2 T_A=125°C Max	Subgroup 3 T_A=-55°C Min	Subgroup 3 T_A=-55°C Max	Unit
		Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16								
V_{IC} (pos)		1	1 mA															GND	M		1.5					Vdc
		2		1 mA															N							
		3							1 mA										CLK							
		4									1 mA								G_1							
		5										1 mA							G_2							
		6											1 mA						D_4							
		7												1 mA					D_3							
		8													1 mA				D_2							
		9														1 mA			D_1							
		10															1 mA		R							
V_{IC} (neg)		11	-1 mA							GND									M		-6.0					
		12		-1 mA															N							
		13							-1 mA										CLK							
		14									-1 mA								G_1							
		15										-1 mA							G_2							
		16											-1 mA						D_4							
		17												-1 mA					D_3							
		18													-1 mA				D_2							
		19														-1 mA			D_1							
		20															-1 mA		R							
V_{OH1}	3007	21	GND	GND					13/		GND	GND	15.0 V	15.0 V	15.0 V	15.0 V	GND	15.0 V	Q_1	14.95		14.95		14.95		
		22																	Q_2							
		23																	Q_3							
		24																	Q_4							
V_{OL1}	3006	25							GND				GND	GND	GND	GND	15.0 V		Q_1		0.05		0.05		0.05	
		26																	Q_2							
		27																	Q_3							
		28																	Q_4							
I_{SS} 2/	3005	29											18.0 V	18.0 V	18.0 V	18.0 V	18.0 V	18.0 V	V_{SS}		-0.25		-2.25			µA
		30															GND									
		31															GND									
		32	18.0 V	18.0 V					18.0 V		18.0 V	18.0V					GND									
		33	18.0 V	18.0 V													18.0 V									
I_{OC11}		34	GND	18.0 V	18.0 V	18.0 V	18.0 V	18.0 V											All outputs together		100					nA
I_{OC21}		35	18.0 V	GND	GND	GND	GND	GND											All outputs together		-100					
I_{OC12}		36	GND	18.0 V	18.0 V														Q_1		25		90		25	
		37				18.0 V													Q_2							
		38					18.0 V												Q_3							
		39						18.0 V											Q_4							
I_{OC22}		40	18.0 V	GND	GND														Q_1		-25		-90		-25	
		41				GND													Q_2							
		42					GND												Q_3							
		43						GND											Q_4							
V_{IH1}		44	1.5 V	1.5 V					14/		1.5 V	1.5 V	3.5 V	1.5 V	1.5 V	1.5 V	1.5 V	5.0 V	Q_4	4.5		4.5		4.5		Vdc
		45												3.5 V	1.5 V	1.5 V			Q_3							
		46									3.5 V		1.5 V						Q_2							
		47										3.5 V		1.5 V	1.5 V	1.5 V			Q_1							
		48									1.5 V		3.5 V						Q_4		0.5		0.5		0.5	
		49										1.5 V		3.5 V	3.5 V	3.5 V	3.5 V		Q_3							
		50																	Q_2							
		51																	Q_1							
		52																	Q_4							
		53																	Q_3							
		54																	Q_2							
		55																	Q_1							
		56																	Q_4							
		57																	Q_3							
		58																	Q_2							
		59																	Q_1							

See footnotes at end of device type 01.

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TABLE III. Group A inspection for device type 01 - Continued.



Symbol	MIL-STD-883 Method	Cases E, F, 2 Test no.	M 1	N 2	Q_1 3	Q_2 4	Q_3 5	Q_4 6	CLK 7	V_{SS} 8	$\overline{Q}_1$ 9	D_2 10	D_4 11	D_3 12	D_2 13	D_1 14	R 15	V_{DD} 16	Measured terminal	Subgroup 1 T_A=25°C Min	Subgroup 1 T_A=25°C Max	Subgroup 2 T_A=125°C Min	Subgroup 2 T_A=125°C Max	Subgroup 3 T_A=-55°C Min	Subgroup 3 T_A=-55°C Max	Unit
			Terminal conditions 1/																Test limits							
V_{IL1}		60	1.5 V	1.5 V					1/	GND	1.5 V	1.5 V	3.5 V	3.5 V	3.5 V	3.5 V	1.5 V	5.0 V	Q_4	4.5	0.5	4.5	0.5	4.5	0.5	Vdd
		61																	Q_3							
		62																	Q_2							
		63																	Q_1							
		64											1.5 V						Q_4							
		65												1.5 V					Q_3							
		66												1.5 V	1.5 V				Q_2							
		67											3.5 V			1.5 V			Q_1							
		68												3.5 V	3.5 V				Q_4							
		69																	Q_3							
		70																	Q_2							
		71																	Q_1							
		72											1.5 V						Q_4							
		73												1.5 V					Q_3							
		74												1.5 V	1.5 V				Q_2							
		75											3.5 V			1.5 V			Q_1							
		76												3.5 V	3.5 V				Q_4							
		77																	Q_3							
		78																	Q_2							
		79																	Q_1							
V_{IH2}		80	3.0 V	3.0 V					1/	GND	3.0 V	3.0 V	3.0 V	3.0 V	3.0 V	3.0 V	3.0 V	10.0 V	Q_4	9.0	1.0	9.0	1.0	9.0	1.0	Vdd
		81																	Q_3							
		82																	Q_2							
		83																	Q_1							
		84									17.0 V		3.0 V						Q_4							
		85												3.0 V					Q_3							
		86												3.0 V	17.0 V				Q_2							
		87											17.0 V			3.0 V			Q_1							
		88												17.0 V	17.0 V				Q_4							
		89																	Q_3							
		90																	Q_2							
		91																	Q_1							
		92											3.0 V						Q_4							
		93												3.0 V					Q_3							
		94												3.0 V	3.0 V				Q_2							
		95											17.0 V			17.0 V			Q_1							
V_{IL2}		96	3.0 V	3.0 V					1/	GND	3.0 V	3.0 V	3.0 V	3.0 V	3.0 V	3.0 V	3.0 V	10.0 V	Q_4	9.0	1.0	9.0	1.0	9.0	1.0	Vdd
		97																	Q_3							
		98																	Q_2							
		99																	Q_1							
		100									17.0 V		3.0 V						Q_4							
		101												3.0 V					Q_3							
		102												3.0 V	17.0 V				Q_2							
		103											17.0 V			3.0 V			Q_1							
		104												17.0 V	17.0 V				Q_4							
		105																	Q_3							
		106																	Q_2							
		107																	Q_1							
		108											3.0 V						Q_4							
		109												3.0 V					Q_3							
		110												3.0 V	3.0 V				Q_2							
		111											17.0 V			17.0 V			Q_1							
		112												17.0 V	17.0 V				Q_4							
		113																	Q_3							
		114																	Q_2							
		115																	Q_1							
		116											3.0 V						Q_4							
		117												3.0 V					Q_3							
		118												3.0 V	3.0 V				Q_2							
		119											17.0 V			17.0 V			Q_1							

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TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD-883 method	Cases E, F, Z	Terminal conditions 1/ M	N	Q_1	Q_2	Q_3	Q_4	CLK	V_{SS}	G_1	G_2	D_4	D_3	D_2	D_1	R	V_{DD}	Measured terminal	Test limits Subgroup 1 T_A=25°C Min	Max	Subgroup 2 T_A=125°C Min	Max	Subgroup 3 T_A=-55°C Min	Max	Unit
		Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16								
V_{IH3}		120-135	4.0 V	4.0 V					16/	GND	4.0 V / 11.0 V / 4.0 V	4.0 V / 11.0 V / 4.0 V	11.0 V / 4.0 V / 11.0 V	4.0 V / 11.0 V / 4.0 V	4.0 V / 11.0 V / 4.0 V	4.0 V / 11.0 V / 4.0 V	4.0 V / 11.0 V	15.0 V	Q_4 Q_3 Q_2 Q_1 $\overline{Q}_4$ $\overline{Q}_3$ $\overline{Q}_2$ $\overline{Q}_1$ Q_4 Q_3 Q_2 Q_1 $\overline{Q}_4$ $\overline{Q}_3$ $\overline{Q}_2$ $\overline{Q}_1$	13.5	1.5	13.5	1.5	13.5	1.5	Vdc
V_{IL3}		136-159											4.0 V / 11.0 V / 4.0 V	4.0 V / 11.0 V / 4.0 V	4.0 V / 11.0 V / 4.0 V	4.0 V / 11.0 V / 4.0 V	4.0 V		Q_4 Q_3 Q_2 Q_1 $\overline{Q}_4$ $\overline{Q}_3$ $\overline{Q}_2$ $\overline{Q}_1$ Q_4 Q_3 Q_2 Q_1 $\overline{Q}_4$ $\overline{Q}_3$ $\overline{Q}_2$ $\overline{Q}_1$	13.5	1.5	13.5	1.5	13.5	1.5	
I_{IH1} 2/	3010	160	18.0 V	18.0 V					18.0 V		18.0 V	18.0 V	18.0 V	18.0 V	18.0 V	18.0 V	18.0 V	18.0 V	All outputs together		10					nA
I_{IH2}		161-170	18.0 V	GND / 18.0 V / GND					GND / GND / 18.0 V / GND		GND / GND / GND / 18.0 V / GND	18.0 V / GND	18.0 V / GND	18.0 V / GND	18.0 V / GND / GND	18.0 V / GND	18.0 V		M W CLK G_1 G_2 D_4 D_3 D_2 D_1 R		1		45			
I_{IL1} 2/	3009	171													GND	GND	GND		All outputs together		-10					
I_{IL2}		172-181	18.0 V	18.0 V / GND / 18.0 V					18.0 V / 18.0 V / GND / 18.0 V		18.0 V / 18.0 V / 18.0 V / GND / 18.0 V	18.0 V / GND / 18.0 V	18.0 V / GND / 18.0 V	18.0 V / GND / 18.0 V	18.0 V / GND / 18.0 V / 18.0 V	18.0 V / GND / 18.0 V	18.0 V / GND		M W CLK G_1 G_2 D_4 D_3 D_2 D_1 R		-1		-45			

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TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD-883 method	Cases E, F, Z	Terminal conditions 1/																Measured terminal	Test limits						Unit
			M	N	Q_1	Q_2	Q_3	Q_4	CLK	V_{SS}	G_1	G_2	D_4	D_3	D_2	D_1	R	V_{DD}		Subgroup 1 T_A=25°C		Subgroup 2 T_A=125°C		Subgroup 3 T_A=-55°C		
		Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Min	Max	Min	Max	Min	Max	
I_{OL1}		182	GND	GND	0.4 V				GND	GND	GND	GND	GND	GND	GND	GND	5.0 V	5.0 V	Q_1	0.51		0.36		0.64		mA
		183				0.4 V													Q_2							
		184					0.4 V												Q_3							
		185						0.4 V											Q_4							
I_{OL2}		186			1.5 V												15.0 V	15.0 V	Q_1	3.4		2.4		4.2		
		187				1.5 V													Q_2							
		188					1.5 V												Q_3							
		189						1.5 V											Q_4							
I_{OH1}		190			4.6 V				17/				5.0 V	5.0 V	5.0 V	5.0 V	GND	5.0 V	Q_1	-0.51		-0.36		-0.64		
		191				4.6 V													Q_2							
		192					4.6 V												Q_3							
		193						4.6 V											Q_4							
I_{OH2}		194			13.5 V				13/				15.0 V	15.0 V	15.0 V	15.0 V		15.0 V	Q_1	-3.4		-2.4		-4.2		
		195				13.5 V													Q_2							
		196					13.5 V												Q_3							
		197						13.5 V											Q_4							
																				Subgroup 4 T_A=25°C						
																				Min	Max					
C_I	3012	198	6/							GND								GND	M		15					pF
		199		6/															N							
		200							6/										CLK							
		201									6/								G_1							
		202										6/							G_2							
		203											6/						D_4		7.5					
		204												6/					D_3							
		205													6/				D_2							
		206														6/			D_1							
		207															6/		R		15					
																				Subgroup 7 T_A=25°C		Subgroup 8 T_A=125°C		Subgroup 8 T_A=-55°C		
																				Min	Max	Min	Max	Min	Max	
Truth table test	3014	208	GND	GND	L	L	L	L	GND	GND	GND	GND	5.0 V	5.0 V	5.0 V	5.0 V	5.0 V	5.0 V	All outputs	See 4/ 5/						
		209			L	L	L	L	GND		GND		5.0 V	5.0 V	5.0 V	5.0 V	GND									
		210			H	H	H	H	5.0 V		GND		5.0 V	5.0 V	5.0 V	5.0 V										
		211							GND		5.0 V		GND	GND	GND	GND										
		212							5.0 V		5.0 V															
		213							GND		GND	5.0 V														
		214							5.0 V		GND															
		215							GND		5.0 V															
		216							5.0 V		5.0 V															
		217							GND		GND	GND														
		218			L	L	L	L	5.0 V																	
		219							GND																	
		220							GND				5.0 V	5.0 V	5.0 V	5.0 V	5.0 V									
		221							5.0 V								5.0 V									
		222							GND		5.0 V	5.0 V					GND									
		223							5.0 V		5.0 V	5.0 V														
		224							GND		GND	GND														
		225			H	H	H	H	5.0 V																	
		226			L	L	L	L	GND																	
		227			L	L	L	L	GND					GND		GND										
		228			H	H	H	H	5.0 V					GND		GND										
		229			H	H	H	H	GND				GND	5.0 V	GND	5.0 V										
		230			H	L	H	L	5.0 V				GND	5.0 V	GND	5.0 V										

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TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD-883 method	Cases E, F, X	Terminal conditions 1/																Measured terminal	Test limits						Unit
			M	N	Q_1	Q_2	Q_3	Q_4	CLK	V_{SS}	Q_1	Q_2	D_4	D_3	D_2	D_1	R	V_{DD}		Subgroup 9 T_A=25°C		Subgroup 10 T_A=125°C		Subgroup 11 T_A=-55°C		
		Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Min	Max	Min	Max	Min	Max	
t_{PHL1}	3003 Fig. 6	231	GND	GND	OUT	OUT	OUT	OUT	IN_1	GND	GND	GND	GND	GND	GND	IN_2	GND	5.0 V	CLK to Q_1	30	600	42	840	30	600	ns
		232											GND	GND	IN_2	GND			CLK to Q_2							
		233											GND	IN_2	GND	GND			CLK to Q_3							
		234											IN_2	GND	GND	GND			CLK to Q_4							
t_{PLH1}		235			OUT	OUT	OUT	OUT					GND	GND	GND	IN_2			CLK to Q_1							
		236											GND	GND	IN_2	GND			CLK to Q_2							
		237											GND	IN_2	GND	GND			CLK to Q_3							
		238											IN_2	GND	GND	GND			CLK to Q_4							
t_{PHL2}		239			OUT	OUT	OUT	OUT					5.0 V	5.0 V	5.0 V	5.0 V	IN_2		R to Q_1	23	460	33	645	23	460	
		240																	R to Q_2							
		241																	R to Q_3							
		242																	R to Q_4							
t_{THL}	3004 Fig. 6	243			OUT	OUT	OUT	OUT							IN_2	IN_2	GND		Q_1	10	200	14	280	10	200	µs
		244												IN_2	5.0 V	5.0 V			Q_2							
		245												5.0 V	5.0 V	5.0 V			Q_3							
		246											IN_2						Q_4							
t_{TLH}		247			OUT	OUT	OUT	OUT					5.0 V	5.0 V	5.0 V	IN_2			Q_1							
		248											5.0 V	5.0 V	IN_2	5.0 V			Q_2							
		249											5.0 V	IN_2	5.0 V				Q_3							
		250											IN_2	5.0 V					Q_4							
t_{PHZ} 7/	Fig. 6	251	IN_2		OUT	OUT	OUT	OUT					5.0 V						M to Q_1	15	300	21	420	15	300	ns
		252																	M to Q_2							
		253																	M to Q_3							
		254																	M to Q_4							
		255	GND	IN_2	OUT														N to Q_1							
		256				OUT	OUT	OUT											N to Q_2							
		257																	N to Q_3							
		258																	N to Q_4							
t_{PLZ} 8/		259	IN_2	GND	OUT	OUT	OUT	OUT					GND	GND	GND	GND			M to Q_1							
		260																	M to Q_2							
		261																	M to Q_3							
		262																	M to Q_4							
		263	GND	IN_2	OUT														N to Q_1							
		264				OUT	OUT	OUT											N to Q_2							
		265																	N to Q_3							
		266																	N to Q_4							
t_{PZH}		267	IN_2	GND	OUT	OUT	OUT	OUT					5.0 V	5.0 V	5.0 V	5.0 V			M to Q_1							
		268																	M to Q_2							
		269																	M to Q_3							
		270																	M to Q_4							
t_{PZL} 8/		271	GND	IN_2	OUT	OUT	OUT	OUT					GND	GND	GND	GND			N to Q_1							
		272																	N to Q_2							
		273																	N to Q_3							
		274																	N to Q_4							
t_{TLH} (CLK) 11/		275		GND	OUT	OUT	OUT	OUT					IN_2	IN_2	IN_2	IN_2			CLK to Q_1		15		15		15	µs
		276																	CLK to Q_2							
		277																	CLK to Q_3							
		278																	CLK to Q_4							
t_{THL} (CLK) 11/		279			OUT	OUT	OUT	OUT											CLK to Q_1							
		280																	CLK to Q_2							
		281																	CLK to Q_3							
		282																	CLK to Q_4							
$t_{P(R)}$ 10/		283			OUT	OUT	OUT	OUT					5.0 V	5.0 V	5.0 V	5.0 V	IN_2		R to Q_1		120		170		120	ns
		284																	R to Q_2							
		285																	R to Q_3							
		286																	R to Q_4							

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TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD-883 method	Cases E, F, Z Test no.	Terminal conditions 1/ M 1	N 2	Q_1 3	Q_2 4	Q_3 5	Q_4 6	CLK 7	V_{SS} 8	G_1 9	G_2 10	D_4 11	D_3 12	D_2 13	D_1 14	R 15	V_{DD} 16	Measured terminal	Test limits Subgroup 9 T_A=25°C Min	Subgroup 9 T_A=25°C Max	Subgroup 10 T_A=125°C Min	Subgroup 10 T_A=125°C Max	Subgroup 11 T_A=-55°C Min	Subgroup 11 T_A=-55°C Max	Unit
t_{SKL}	Fig. 6	287	GND	GND	OUT				IN_1	GND	GND	GND	5.0 V	5.0 V	5.0 V	IN_2	GND	5.0 V	CLK to D_1		200		280		200	ns
		288				OUT							5.0 V	5.0 V	IN_2	5.0 V			CLK to D_2							
		289					OUT						5.0 V	IN_2	5.0 V	5.0 V			CLK to D_3							
		290						OUT					IN_2	5.0 V	5.0 V	5.0 V			CLK to D_4							
t_{SHL}		291			OUT								5.0 V	5.0 V	5.0 V	IN_2			D_1 to CLK							
		292				OUT							5.0 V	5.0 V	IN_2	5.0 V			D_2 to CLK							
		293					OUT						5.0 V	IN_2	5.0 V				D_3 to CLK							
		294						OUT					IN_2	5.0 V					D_4 to CLK							
f_{CLK} 9/		295			OUT								5.0 V						CLK to Q_1		670		1,000		670	
		296				OUT													CLK to Q_2							
		297					OUT												CLK to Q_3							
		298						OUT											CLK to Q_4							
$t_{D(CLK)}$ 10/		299			OUT														CLK to Q_1		200		280		200	
		300				OUT													CLK to Q_2							
		301					OUT												CLK to Q_3							
		302						OUT											CLK to Q_4							
																				Subgroup 12 T_A=25°C Min	Subgroup 12 T_A=25°C Max					
t_{PHL1}	3003 Fig. 6	303	GND	GND	OUT				IN_1	GND	GND	GND	GND	GND	GND	IN_2	GND	10.0 V	CLK to Q_1	13	250					ns
		304				OUT							GND	GND	IN_2	GND			CLK to Q_2							
		305					OUT						GND	IN_2	GND	GND			CLK to Q_3							
		306						OUT					IN_2	GND	GND	GND			CLK to Q_4							
t_{PLH1}		307			OUT								GND	GND	GND	IN_2			CLK to Q_1							
		308				OUT							GND	GND	IN_2	GND			CLK to Q_2							
		309					OUT						GND	IN_2	GND	GND			CLK to Q_3							
		310						OUT					IN_2	GND	GND	GND			CLK to Q_4							
t_{PHL2}		311			OUT								10.0 V	10.0 V	10.0 V	10.0 V	IN_2		R to Q_1	10	200					
		312				OUT													R to Q_2							
		313					OUT												R to Q_3							
		314						OUT											R to Q_4							
t_{THL}	3004 Fig. 6	315			OUT											IN_2	GND		Q_1	5	100					
		316				OUT									IN_2	10.0 V			Q_2							
		317					OUT							IN_2	10.0 V	10.0 V			Q_3							
		318						OUT					IN_2	10.0 V	10.0 V	10.0 V			Q_4							
t_{TLH}		319			OUT								10.0 V	10.0 V	10.0 V	IN_2			Q_1							
		320				OUT							10.0 V	10.0 V	IN_2	10.0 V			Q_2							
		321					OUT						10.0 V	IN_2	10.0 V				Q_3							
		322						OUT					IN_2	10.0 V	10.0 V				Q_4							
t_{PHZ} 7/	Fig. 6	323	IN_2		OUT								10.0 V						M to Q_1	8	150					
		324				OUT													M to Q_2							
		325					OUT												M to Q_3							
		326						OUT											M to Q_4							
		327	GND	IN_2	OUT														N to Q_1							
		328				OUT													N to Q_2							
		329					OUT												N to Q_3							
		330						OUT											N to Q_4							
t_{PLZ} 8/		331	IN_2	GND	OUT								GND	GND	GND	GND			M to Q_1							
		332				OUT													M to Q_2							
		333					OUT												M to Q_3							
		334						OUT											M to Q_4							
		335	GND	IN_2	OUT														N to Q_1							
		336				OUT													N to Q_2							
		337					OUT												N to Q_3							
		338						OUT											N to Q_4							

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TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD-883 method	Cases E, F, Z	Terminal conditions 1/ H	N	Q_1	Q_2	Q_3	Q_4	CLK	V_{SS}	G_1	G_2	D_4	D_3	D_2	D_1	R	V_{DD}	Measured terminal	Test limits Subgroup 12 T_A=25°C Min	Max	Unit
		Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16				
t_{PZH} 7/	Fig. 6	339	IN_2	GND	OUT	OUT			IN_1	GND	GND	GND	10.0 V	10.0 V	10.0 V	10.0 V	GND	10.0 V	M to Q_1	8	150	ns
		340					OUT												M to Q_2			
		341						OUT											M to Q_3			
		342																	M to Q_4			
t_{PZL} 8/		343	GND	IN_2	OUT	OUT							GND	GND	GND	GND			N to Q_1			
		344					OUT												N to Q_2			
		345						OUT											N to Q_3			
		346																	N to Q_4			

1/ Pins not designated may be "high" level logic, "low" level logic or open. Exceptions are as follows: $V_{IC(pos)}$ tests, the V_{SS} terminals shall be open; $V_{IC(neg)}$ tests, the V_{DD} terminal shall be open; I_{SS} tests, the output terminals shall be open.
2/ The I_{SS} measurements shall be performed in sequence.
3/ The device manufacturer may, at his option, measure I_{IL} and I_{IH} at 25°C for each individual input or measure all inputs together.
4/ The truth table tests shall be performed in sequence.
5/ The truth table tests shall be performed at V_{IH} and V_{DD} ≤5 Vdc and ≥18 Vdc. "L" = V_{SS} +0.5 V maximum and "H" = V_{DD} -0.5 V minimum.
6/ See 4.4.1c.
7/ Pins 3, 4, 5, and 6 tied to V_{SS} through 1 Kohm resistors.
8/ Pins 3, 4, 5, and 6 tied to V_{DD} through 1 Kohm resistors.
9/ The minimum clock frequency (f_{CLK}) requirement is considered met if proper output state changes occur with the pulse repetition period set to that given in the limits column.
10/ The minimum clock or reset pulse width ($t_{p(CLK)}$, $t_{p(R)}$) requirement is considered met if proper output state changes occur with the pulse width set to that given in the limits column.
11/ Pulse repetition period = 100 µs, 50 percent duty cycle. The maximum clock transition time $t_{TLH(CLK)}$, $t_{THL(CLK)}$ requirement is considered met if proper output state changes occur with the rise time set to that given in the limits column.
12/ 100 Kohm resistors shall be connected between pins 2 and 14 to GND.
13/ Apply clock pulse; V_{IN} = 0 to 15 Vdc.
14/ Apply clock pulse; V_{IN} = 1.5 to 3.5 Vdc.
15/ Apply clock pulse; V_{IN} = 3.0 to 7.0 Vdc.
16/ Apply clock pulse; V_{IN} = 4.0 to 11.0 Vdc.
17/ Apply clock pulse; V_{IN} = 0 to 5 Vdc.

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TABLE III. Group A inspection for device type 02.

Symbol	MIL-STD-883 method	Cases A,C,D,X,Y Symbol	Terminal conditions 1/ NC	R	J_1	J_2	J_3	$\overline{Q}$	V_{SS}	Q	K_3	K_2	K_1	CLK	S	V_{DD}	Measured terminal	Test limits Subgroup 1 T_A = 25°C Min	Max	Subgroup 2 T_A = 125°C Min	Max	Subgroup 3 T_A = -55°C Min	Max	Units
		Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14								
V_{IC} (pos)		1		1 mA												GND	R		1.5					Vdc
		2			1 mA												J_1							
		3				1 mA											J_2							
		4					1 mA										J_3							
		5									1 mA						K_3							
		6										1 mA					K_2							
		7											1 mA				K_1							
		8												1 mA			CLK							
		9													1 mA		S							
V_{IC} (neg)		10		-1 mA					GND								R		-6.0					
		11			-1 mA												J_1							
		12				-1 mA											J_2							
		13					-1 mA										J_3							
		14									-1 mA						K_3							
		15										-1 mA					K_2							
		16											-1 mA				K_1							
		17												-1 mA			CLK							
		18													-1 mA		S							
I_{SS} 2/	3005	19		GND	18.0 V	18.0 V	18.0 V				18.0 V	18.0 V	18.0 V	GND	18.0 V	18.0 V	V_{SS}		-.075		-2.5			µA
		20		18.0 V	18.0 V	18.0 V	18.0 V				18.0 V	18.0	18.0 V	GND	GND									
		21		GND	18.0 V	18.0 V	18.0 V				18.0 V	18.0	18.0 V	18.0 V										
		22			GND	GND	GND				GND	GND	GND	GND										
V_{OH1}	3006	23			15.0 V	15.0 V	15.0 V				15.0 V	15.0 V	15.0 V	10/		15.0 V	Q	14.95		14.95		14.95		Vdc
V_{OL1}		24												GND			$\overline{Q}$		0.05		0.05		0.05	
V_{OL1}		25												10/			Q		0.05		0.05		0.05	
V_{OH1}		26												GND			$\overline{Q}$	14.95		14.95		14.95		

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TABLE III. Group A inspection for device type 02 -Continued.

Symbol	MIL-STD-883 method	Cases A,C,D,X,Y	Terminal conditions 1/													Measured terminal	Test limits						Units
																	Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Symbol	NC	R	J_1	J_2	J_3	$\overline{Q}$	V_{SS}	Q	K_3	K_2	K_1	CLK	S	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14								
V_{IL1} V_{IL2} V_{IL3}		27 28 29		12/	12/	12/	12/	12/	GND	12/	12/	12/	12/	12/	12/	12/	All outputs	12/	12/	12/	12/	12/	12/	
V_{IH1} V_{IH2} V_{IH3}		30 31 32																						
I_{OH1} I_{OH1}		33 34		GND	5.0 V	5.0 V	5.0 V	4.6 V		4.6 V	5.0 V	5.0 V	5.0 V	11/ 11/	GND	5.0 V	Q $\overline{Q}$	-0.51 -0.51		-0.36 -0.36		-0.64 -0.64		mA
I_{OL1} I_{OL1}		35 36						0.4 V		0.4 V				GND 11/			Q $\overline{Q}$	0.51 0.51		0.36 0.36		0.64 0.64		
I_{OH2} I_{OH2}		37 38			15.0 V	15.0 V	15.0 V	13.5 V		13.5 V	15.0 V	15.0 V	15.0 V	10/ 10/		15.0 V	Q $\overline{Q}$	-3.4 -3.4		-2.4 -2.4		-4.2 -4.2		
I_{OL2} I_{OL2}		39 40						1.5 V		1.5 V				GND 10/			Q $\overline{Q}$	3.4 3.4		2.4 2.4		4.2 4.2		
I_{IH1} 3/	3010	41		18.0 V	18.0 V	18.0 V	18.0 V				18.0 V	18.0 V	18.0 V	18.0 V	18.0 V	18.0 V	All inputs together		9					nA
I_{IH2}		42 43 44 45 46 47 48 49 50		18.0 V GND	GND 18.0 V GND	GND GND 18.0 V GND	GND GND GND 18.0 V GND				GND 18.0 V GND	GND 18.0 V GND GND GND	GND GND 18.0 V GND GND	GND GND GND 18.0 V GND	GND 18.0 V	GND GND	R J_1 J_2 J_3 K_3 K_2 K_1 CLK S		1		45			

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TABLE III. Group A inspection for device type 02 -Continued.

Symbol	MIL-STD-883 method	Cases A,C,D,X,Y	Terminal conditions 1/														Measured terminal	Test limits						Units
																		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Symbol	NC	R	J_1	J_2	J_3	$\overline{Q}$	V_{SS}	Q	K_3	K_2	K_1	CLK	S	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14								
I_{IL1} 3/	3009	51		GND	GND	GND	GND		GND		GND	GND	GND	GND	GND	18.0 V	All inputs together		-9					nA
I_{IL2}		52		GND	18.0 V	18.0 V	18.0 V				18.0 V	18.0 V	18.0 V	18.0 V	18.0 V		R		-1		-45			
		53		18.0 V	GND	18.0 V	18.0 V										J_1							
		54			18.0 V	GND	18.0V										J_2							
		55				18.0 V	GND										J_3							
		56					18.0 V				GND						K_3							
		57									18.0 V	GND					K_2							
		58										18.0 V	GND				K_1							
		59										18.0 V	18.0 V	GND			CLK							
		60										18.0 V	18.0 V	18.0 V	GND		S							
																		Subgroup 4 T_A = 25°C						
																		Min	Max					
C_I	3012	61		4/					GND							GND	R		7.5					pF
		62			4/												J_1							
		63				4/											J_2							
		64					4/										J_3							
		65									4/						K_3							
		66										4/					K_2							
		67											4/				K_1							
		68												4/			CLK							
		69													4/		S							

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TABLE III. Group A inspection for device type 02 -Continued.

Symbol	MIL-STD-883 method	Cases A,C,D,X,Y	Terminal conditions 1/													Measured terminal	Test limits						Units
																	Subgroup 7 T_A = 25°C		Subgroup 8				
																			T_A = 125°C		T_A = -55°C		
		Symbol	NC	R	J_1	J_2	J_3	$\overline{Q}$	V_{SS}	Q	K_3	K_2	K_1	CLK	S	V_{DD}							
		Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14		Min	Max	Min	Max	Min	Max	
Truth table test	3014	70		GND	5.0 V	5.0 V	5.0 V	L	GND	H	5.0 V	5.0 V	5.0 V	GND	5.0 V	5.0 V	All outputs	See notes 5/, 6/						
		71		5.0 V				H		L				GND	GND									
		72		GND				H		L				GND										
		73						L		H				5.0 V										
		74			GND			L		H				GND										
		75						H		L				5.0 V										
		76						H		L				GND										
		77						H		L				5.0 V										
		78			5.0 V			L		H				GND	5.0 V									
		79				GND		L		H				GND	GND									
		80						H		L				5.0 V										
		81						H		L				GND										
		82						H		L				5.0 V										
		83				5.0 V		L		H				5.0 V	5.0 V									
		84					GND	L		H				GND	GND									
		85						H		L				5.0 V										
		86												GND										
		87											GND	5.0 V										
		88					5.0 V							GND										
		89						L		H				5.0 V										
		90						L		H				GND										
		91						L		H			GND	5.0 V										
		92		5.0 V				H		L			5.0 V	GND										
		93		GND				H		L		GND		GND										
		94						L		H				5.0 V										
		95						L		H				GND										
		96						L		H				5.0 V										

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TABLE III. Group A inspection for device type 02 - Continued.

Symbol	MIL-STD-883 method	Cases A,C,D,X,Y	Terminal conditions 1/																Measured terminal	Test limits						Units
																				Subgroup 7 T_A = 25°C		Subgroup 8 T_A = 125°C		Subgroup 8 T_A = -55°C		
		Test No.	Symbol	NC	R	J_1	J_2	J_3	$\overline{Q}$	V_{SS}	Q	K_3	K_2	K_1	CLK	S	V_{DD}			Min	Max	Min	Max	Min	Max	
				1	2	3	4	5	6	7	8	9	10	11	12	13	14									
Truth table test	3014	97			5.0 V	5.0 V	5.0 V	5.0 V	H	GND	L	GND	5.0 V	5.0 V	GND	GND	5.0 V	All outputs	See notes 5/, 6/						
		98			GND				H		L															
		99							L		H				5.0 V											
		100				GND	GND	GND							GND											
		101													5.0 V											
		102				GND	GND	GND					GND	GND	GND											
		103													5.0 V											
		104													GND											
		105													5.0 V											
t_{PHL1}	3003 Fig.6	106			GND	5.0 V	5.0 V	5.0 V		GND		5.0 V	5.0 V	5.0 V	IN1	GND	5.0 V	CLK to Q		25	500	35	700	25	500	ns
t_{PHL1}		107							OUT		OUT							CLK to $\overline{Q}$								
t_{PLH1}		108									OUT							CLK to Q								
t_{PLH1}		109				GND	GND	GND				GND	GND	GND		IN2		CLK to $\overline{Q}$		15	300	21	420	15	300	
t_{PHL2}		110			IN1				OUT		OUT							S to Q								
t_{PHL2}		111														IN2		S to $\overline{Q}$								
t_{PLH2}		112									OUT					GND		R to Q								
t_{PLH2}		113				GND	GND	GND				GND	GND	GND				R to $\overline{Q}$								
t_{THL}	3004 Fig.6	114			GND				OUT		OUT				IN1			CLK to Q		10	200	14	280	10	200	
t_{THL}		115				5.0 V	5.0 V	5.0 V				5.0 V	5.0 V	5.0 V				CLK to $\overline{Q}$								
t_{TLH}		116							OUT		OUT							CLK to Q								
t_{TLH}		117																CLK to $\overline{Q}$								
f_{CLK} 2/		118							OUT		OUT							CLK to Q			1000		1400		1000	
f_{CLK} 2/		119																CLK to $\overline{Q}$			1000		1400		1000	

	Subgroup 9 T_A = 25°C		Subgroup 10 T_A = 125°C		Subgroup 11 T_A = -55°C	
	Min	Max	Min	Max	Min	Max
	25	500	35	700	25	500
	15	300	21	420	15	300
	10	200	14	280	10	200
		1000		1400		1000
		1000		1400		1000

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TABLE III. Group A inspection for device type 02 -Continued.

Symbol	MIL-STD-883 method	Cases A,C,D,X,Y	Terminal conditions 1/														Measured terminal	Test limits						Units
																		Subgroup 9 T_A = 25°C		Subgroup 10 T_A = 125°C		Subgroup 11 T_A = -55°C		
		Symbol	NC	R	J_1	J_2	J_3	$\overline{Q}$	V_{SS}	Q	K_3	K_2	K_1	CLK	S	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14								
t_P(CLK) 8/	3004 Fig. 6	120		GND	5.0 V	5.0 V	5.0 V		GND	OUT	5.0 V	5.0 V	5.0 V	IN_1	GND	5.0 V	CLK to Q		140		200		140	
t_P(CLK) 8/		121						OUT							GND		CLK to $\overline{Q}$		140		200		140	
t_{PS} 8/		122								OUT					IN_2		S to Q		200		280		200	
t_{PS} 8/		123						OUT							IN_2		S to $\overline{Q}$							
t_{PR} 8/		124		IN_2						OUT					GND		R to Q							
t_{PR} 8/		125		IN_2				OUT									R to $\overline{Q}$							
t_{TLH}(CLK) 9/		126		GND						OUT							CLK to Q		15		15		15	us
t_{TLH}(CLK) 9/		127						OUT									CLK to $\overline{Q}$							
t_{THL}(CLK) 9/		128								OUT							CLK to Q							
t_{THL}(CLK) 9/		129						OUT									CLK to $\overline{Q}$							
t_{SHL1}		130			IN_2					OUT							CLK to J_1		400		560		400	ns
		131			5.0 V	IN_2											CLK to J_2							
		132				5.0 V	IN_2										CLK to J_3							
		133					5.0 V				IN_2						CLK to K_3							
		134									5.0 V	IN_2					CLK to K_2							
		135										5.0 V	IN_2				CLK to K_1							
t_{SLH1}		136			IN_2								5.0 V				J_1 to CLK							
		137			5.0 V	IN_2											J_2 to CLK							
		138				5.0 V	IN_2										J_3 to CLK							
		139					5.0 V				IN_2						K_3 to CLK							
		140									5.0 V	IN_2					K_2 to CLK							
		141									5.0 V	5.0 V	IN_2				K_1 to CLK							

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TABLE III. Group A inspection for device type 02 -Continued.

Symbol	MIL-STD-883 method	Cases A,C,D,X,Y	Terminal conditions 1/														Measured terminal	Test limits						Units
		Symbol	NC	R	J_1	J_2	J_3	$\overline{Q}$	V_{SS}	Q	K_3	K_2	K_1	CLK	S	V_{DD}		Subgroup 12 T_A = 25°C					
		Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14		Min	Max	Min	Max	Min	Max	
t_{PHL1}	3003 Fig.6	142		GND	10.0 V	10.0 V	10.0 V		GND	OUT	10.0 V	10.0 V	10.0 V	IN_1	GND	10.0 V	CLK to Q	13	250					ns
t_{PHL1}		143						OUT									CLK to $\overline{Q}$							
t_{PLH1}		144								OUT							CLK to Q							
t_{PLH1}		145						OUT									CLK to $\overline{Q}$							
t_{PHL2}		146		IN_1	GND	GND	GND			OUT	GND	GND	GND	GND	IN_2		S to Q	8	150					
t_{PHL2}		147						OUT							IN_2		S to $\overline{Q}$							
t_{PLH2}		148								OUT					GND		R to Q							
t_{PLH2}		149						OUT									R to $\overline{Q}$							
t_{THL}	3004 Fig.6	150		GND	10.0 V	10.0 V	10.0 V			OUT	10.0 V	10.0 V	10.0 V	IN_1			Q	5	100					
t_{THL}		151						OUT									$\overline{Q}$							
t_{TLH}		152								OUT							Q							
t_{TLH}		153						OUT									$\overline{Q}$							

1/ Pins not designated may be "high" level logic, "low" level logic or open. Exceptions are as follows: $V_{IC(pos)}$ tests, the V_{SS} terminals shall be open; $V_{IC(neg)}$ tests, the V_{DD} terminal shall be open; I_{SS} tests, the output terminals shall be open.
2/ The I_{SS} measurements shall be performed in sequence.
3/ The device manufacturer may, at his option, measure I_{IL} and I_{IH} at 25°C for each individual input or measure all inputs together.
4/ See 4.4.1c.
5/ The truth table tests shall be performed in sequence.
6/ The truth table tests shall be performed at V_{IH} and V_{DD} $\geq$5 Vdc and $\geq$ 18 Vdc. "L" = V_{SS} + 0.5 V maximum and "H" = V_{DD} -0.5 V minimum.
7/ The minimum clock frequency (f_{CLK}) requirement is considered met if proper output state changes occur with the pulse repetition period set to that given in the limits column.
8/ The minimum clock, reset or set pulse width (t_p(CLK), t_p(R), t_p(S)) requirement is considered met if proper output state changes occur with the pulse width set to that given in the limits column.
9/ Pulse repetition period = 100 µs, 50 percent duty cycle. The maximum clock transition time $t_{TLH(CLK)}$, $t_{THL(CLK)}$ requirement is considered met if proper output state changes occur with rise time set to that given in the limits column.
10/ Apply clock pulse; V_{IN} = 0 to 15 Vdc.
11/ Apply clock pulse; V_{IN} = 0 to 5 Vdc.

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12/ The input/output conditions and timing sequence shall apply:

Input/output conditions					
Test	V_{DD}	Input levels		Output levels	
		1	0	1	0
				Min	Max
V_{IL1} V_{IH}	5.0 V	3.5 V	1.5 V	4.5 V	0.5 V
V_{IL2} V_{IH2}	10.0 V	7.0 V	3.0 V	9.0 V	1.0 V
V_{IL3} V_{IH3}	15.0 V	11.0 V	4.0 V	13.5 V	1.5 V

* Time slot	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36
INPUTS R	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	0	0	0	0
J_1	1	1	1	1	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0
J_2	1	1	1	1	1	1	1	1	1	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0
J_3	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0
K_3	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	1	1	1	1	1	0	0	0	0
K_2	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1	0	0	0	0
K_1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0
CLK	0	0	0	1	0	1	0	1	0	0	1	0	1	0	0	1	0	1	0	1	0	1	0	0	1	0	1	0	0	1	0	1	0	1	0	1
S	1	0	0	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
OUTPUTS $\bar{Q}$	0	1	1	0	0	1	1	1	0	0	1	1	1	0	0	1	1	1	1	0	0	0	1	1	0	0	0	1	1	0	0	0	0	0	0	0
Q	1	0	0	1	1	0	0	0	1	1	0	0	0	1	1	0	0	0	0	1	1	1	0	0	1	1	1	0	0	1	1	1	1	1	1	1

* Tests are to be run in sequence.

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TABLE III. Group A inspection for device type 03.

Symbol	MIL-STD-883 method	Cases A,C,D,X,Y	Terminal conditions 1/														Measured terminal	Test limits						Units
																		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Symbol	NC	R	J_1	J_2	$\overline{J}_3$	$\overline{Q}$	V_{SS}	Q	$\overline{K}_3$	K_2	K_1	CLK	S	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14								
$V_{IC(pos)}$		1		1 mA												GND	R		1.5					Vdc
		2			1 mA												J_1							
		3				1 mA											J_2							
		4					1 mA										$\overline{J}_3$							
		5									1 mA						$\overline{K}_3$							
		6										1 mA					K_2							
		7											1 mA				K_1							
		8												1 mA			CLK							
		9													1 mA		S							
$V_{IC(neg)}$		10		-1 mA					GND								R		-6.0					
		11			-1 mA												J_1							
		12				-1 mA											J_2							
		13					-1 mA										$\overline{J}_3$							
		14									-1 mA						$\overline{K}_3$							
		15										-1 mA					K_2							
		16											-1 mA				K_1							
		17												-1 mA			CLK							
		18													-1 mA		S							
I_{SS} 2/	3005	19		GND	18.0 V	18.0 V	GND				GND	18.0 V	18.0 V	GND	18.0 V	18.0 V	V_{SS}		-.075		-2.5			µA
		20		18.0 V	18.0 V	18.0 V	GND				GND	18.0 V	18.0 V	GND	GND										
		21		GND	18.0 V	18.0 V	GND				GND	18.0 V	18.0 V	18.0 V											
		22			GND	GND	18.0 V				18.0 V	GND	GND	GND											
V_{OH1}	3006	23			15 V	15 V	GND				GND	15 V	15 V	10/		15 V	Q	14.95		14.95		14.95		Vdc
V_{OL1}	3007	24												GND			$\overline{Q}$		0.05		0.05		0.05	
V_{OL1}	3007	25												10/			Q		0.05		0.05		0.05	
V_{OH1}	3006	26												GND			$\overline{Q}$	14.95		14.95		14.95		

See footnotes at end of device type 03.

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TABLE III. Group A inspection for device type 03 -Continued.

Symbol	MIL-STD-883 method	Cases A,C,D,X,Y	Terminal conditions 1/														Measured terminal	Test limits						Units
																		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Symbol	NC	R	J_1	J_2	J_3	$\overline{Q}$	V_{SS}	Q	$\overline{K}_3$	K_2	K_1	CLK	S	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14								
V_{IL1}		27		12/	12/	12/	12/	12/	GND	12/	12/	12/	12/	12/	12/	12/	All outputs	12/	12/	12/	12/	12/	12/	12/
V_{IL2}		28		↓	↓	↓	↓	↓		↓	↓	↓	↓	↓	↓	↓		↓	↓	↓	↓	↓	↓	↓
V_{IL3}		29		↓	↓	↓	↓	↓		↓	↓	↓	↓	↓	↓	↓		↓	↓	↓	↓	↓	↓	↓
V_{IH1}		30		↓	↓	↓	↓	↓		↓	↓	↓	↓	↓	↓	↓		↓	↓	↓	↓	↓	↓	↓
V_{IH2}		31		↓	↓	↓	↓	↓		↓	↓	↓	↓	↓	↓	↓		↓	↓	↓	↓	↓	↓	↓
V_{IH3}		32		↓	↓	↓	↓	↓		↓	↓	↓	↓	↓	↓	↓		↓	↓	↓	↓	↓	↓	↓
I_{OH1}		33		GND	5.0 V	5.0 V	GND			4.6 V	GND	5.0 V	5.0 V	11/	GND	5.0 V	Q	-0.51		-0.36		-0.64		mA
I_{OH1}		34		↓	↓	↓	↓	4.6 V		↓	↓	↓	↓	11/	↓	↓	$\overline{Q}$	-0.51		-0.36		-0.64		↓
I_{OL1}		35								0.4 V				GND			Q	0.51		0.36		0.64		
I_{OL1}		36			↓	↓		0.4 V		↓		↓	↓	11/		↓	$\overline{Q}$	0.51		0.36		0.64		
I_{OH2}		37			15.0 V	15.0 V				13.5 V		15.0 V	15.0 V	10/		15.0 V	Q	-3.4		-2.4		-4.2		
I_{OH2}		38			↓	↓		13.5 V		↓		↓	↓	10/		↓	$\overline{Q}$	-3.4		-2.4		-4.2		
I_{OL2}		39								1.5 V				GND			Q	3.4		2.4		4.2		
I_{OL2}		40		↓	↓	↓	↓	1.5 V		↓	↓	↓	↓	10/	↓	↓	$\overline{Q}$	3.4		2.4		4.2		↓
I_{IH1} 3/	3010	41		18.0 V	18.0 V	18.0 V	18.0 V				18.0 V	18.0 V	18.0 V	18.0 V	18.0 V	18.0 V	All inputs together		9					nA
I_{IH2}		42		18.0 V	GND	GND	GND				GND	GND	GND	GND	GND		R		1		45			
		43		GND	18.0 V	GND	GND				↓	↓	↓	↓	↓		J_1		↓		↓			
		44			GND	18.0 V	GND										J_2							
		45				GND	18.0 V										J_3							
		46					GND				18.0 V						$\overline{K}_3$							
		47									GND	18.0 V					K_2							
		48										GND	18.0 V				K_1							
		49											GND	18.0 V			CLK							
		50		↓	↓	↓	↓				↓	↓	GND	GND	18.0 V		S		↓		↓			
I_{IL1} 3/	3009	51		↓	↓	↓	↓		↓		↓	↓	GND	GND	GND	↓	All inputs together		-9					↓

See footnotes at end of device type 03.

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TABLE III. Group A inspection for device type 03 -Continued.

Symbol	MIL-STD-883 method	Cases A,C,D,X,Y	Terminal conditions 1/															Measured terminal	Test limits						Units
																			Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Symbol	NC	R	J_1	J_2	$\overline{J}_3$	$\overline{Q}$	V_{SS}	Q	$\overline{K}_3$	K_2	K_1	CLK	S	V_{DD}			Min	Max	Min	Max	Min	Max	
		Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14									
I_{IL2}	3009	52		GND	18.0 V	18.0 V	18.0 V		GND		18.0 V	18.0 V	18.0 V	18.0 V	18.0 V	18.0 V	R			-1		-45			nA
		53		18.0 V	GND	18.0 V	18.0 V										J_1								
		54			18.0 V	GND	18.0 V										J_2								
		55				18.0 V	GND										$\overline{J}_3$								
		56					18.0 V				GND						$\overline{K}_3$								
		57									18.0 V	GND					K_2								
		58										18.0 V	GND				K_1								
		59										18.0 V	18.0 V	GND			CLK								
		60										18.0 V	18.0 V	18.0 V	GND		S								
																		Subgroup 4 T_A = 25°C							
																		Min	Max						
C_1	3012	61		6/					GND							GND	R		7.5						pF
		62			6/												J_1								
		63				6/											J_2								
		64					6/										$\overline{J}_3$								
		65									6/						$\overline{K}_3$								
		66										6/					K_2								
		67											6/				K_1								
		68												6/			CLK								
		69													6/		S								
																		Subgroup 7 T_A = 25°C		Subgroup 8					
																				T_A = 125°C		T_A = -55°C			
																		Min	Max	Min	Max	Min	Max		
Truth table test	3014	70		GND	5.0 V	5.0 V	GND	L	GND	H	GND	5.0 V	5.0 V	GND	5.0 V	5.0 V	All outputs	See 4/, 5/							
		71		5.0 V				H		L				GND	GND										
		72		GND				H		L				GND	GND										
		73		GND				L		H				5.0 V											

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TABLE III. Group A inspection for device type 03 -Continued.

Symbol	MIL-STD-883 method	Cases A,C,D,X,Y	Terminal conditions 1/														Measured terminal	Test limits						Units
																		Subgroup 7		Subgroup 8				
		Symbol	NC	R	J_1	J_2	$\overline{J}_3$	$\overline{Q}$	V_{SS}	Q	$\overline{K}_3$	K_2	K_1	CLK	S	V_{DD}		T_A = 25°C		T_A = 125°C		T_A = -55°C		
		Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14		Min	Max	Min	Max	Min	Max	
Truth table test	3014	74		GND	GND	5.0 V	GND	L	GND	H	GND	5.0 V	5.0 V	GND	GND	5.0 V	All outputs	See 4/, 5/						
		75						H		L				5.0 V										
		76						H		L				GND										
		77						H		L				5.0 V										
		78			5.0 V			L		H				GND	5.0 V									
		79				GND		L		H				GND	GND									
		80						H		L				5.0 V										
		81						H		L				GND										
		82						H		L				5.0 V										
		83				5.0 V		L		H				GND	5.0 V									
		84					5.0 V	L		H				GND	GND									
		85						H		L				5.0 V										
		86						H		L				GND										
		87						H		L			GND	5.0 V										
		88					GND	H		L				GND										
		89						L		H			GND	5.0 V										
		90						L		H			5.0 V	GND										
		91						L		H			5.0 V	5.0 V										
		92		5.0 V				H		L				GND										
		93		GND				H		L		GND		GND										
		94						L		H				5.0 V										
		95						L		H				GND										
		96						L		H				5.0 V										
		97		5.0 V				H		L	5.0 V	5.0 V		GND										
		98		GND				H		L				GND										
		99		GND				L		H				5.0 V										
		100		GND				L		H				GND										

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TABLE III. Group A inspection for device type 03 -Continued.

Symbol	MIL-STD-883 method	Cases A,C,D,X,Y	Terminal conditions 1/														Measured terminal	Test limits						Units
																		Subgroup 7 T_A = 25°C		Subgroup 8 T_A = 125°C		T_A = -55°C		
		Symbol	NC	R	J_1	J_2	$\overline{J}_3$	$\overline{Q}$	V_{SS}	Q	$\overline{K}_3$	K_2	K_1	CLK	S	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14								
Truth table test	3014	101		GND	5.0 V	5.0 V	GND	L	GND	H	5.0 V	5.0 V	5.0 V	5.0 V	GND	5.0 V	All outputs	See 4/, 5/						
		102			GND	GND	5.0 V					GND	GND	GND										
		103												5.0 V										
		104												GND										
		105												5.0 V										
																		Subgroup 9 T_A = 25°C		Subgroup 10 T_A = 125°C		Subgroup 11 T_A = -55°C		
																		Min	Max	Min	Max	Min	Max	
t_{PHL1}	3003 Fig. 6	106		GND	5.0 V	5.0 V	GND		GND	OUT	GND	5.0 V	5.0 V	IN_1	GND	5.0 V	CLK to Q	25	500	35	700	25	500	ns
t_{PHL1}		107						OUT									CLK to $\overline{Q}$							
t_{PLH1}		108								OUT							CLK to Q							
t_{PLH1}		109						OUT									CLK to $\overline{Q}$							
t_{PHL2}		110		IN_1	GND	GND	5.0 V			OUT	5.0 V	GND	GND	GND	IN_2		S to Q	15	300	21	420	15	300	
t_{PHL2}		111						OUT							IN_2		S to $\overline{Q}$							
t_{PLH2}		112								OUT					GND		R to Q							
t_{PLH2}		113						OUT									R to $\overline{Q}$							
t_{THL}	3004 Fig. 6	114		GND	5.0 V	5.0 V	GND			OUT	GND	5.0 V	5.0 V	IN_1			CLK to Q	10	200	14	280	10	200	
t_{THL}		115						OUT									CLK to $\overline{Q}$							
t_{TLH}		116								OUT							CLK to Q							
t_{TLH}		117						OUT									CLK to $\overline{Q}$							
f_{CLK} 7/		118								OUT							CLK to Q		1000		1400		1000	
f_{CLK} 7/		119						OUT									CLK to $\overline{Q}$		1000		1400		1000	
$t_{p(CLK)}$ 8/		120								OUT							CLK to Q		140		200		140	
$t_{p(CLK)}$ 8/		121						OUT									CLK to $\overline{Q}$		140		200		140	
$t_{p(S)}$ 8/		122								OUT					IN_2		CLK to Q		200		280		200	
$t_{p(S)}$ 8/		123						OUT							IN_2		CLK to $\overline{Q}$							
$t_{p(R)}$ 8/		124		IN_2						OUT					GND		CLK to Q							
$t_{p(R)}$ 8/		125		IN_2				OUT							GND		CLK to $\overline{Q}$							

See footnotes at end of device type 03.

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TABLE III. Group A inspection for device type 03 -Continued.

Symbol	MIL-STD-883 method	Cases A,C,D,X,Y	Terminal conditions 1/														Measured terminal	Test limits						Units
																		Subgroup 9 T_A = 25°C		Subgroup 10 T_A = 125°C		Subgroup 11 T_A = -55°C		
		Symbol	NC	R	J_1	J_2	$\overline{J}_3$	$\overline{Q}$	V_{SS}	Q	$\overline{K}_3$	K_2	K_1	CLK	S	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14								
$t_{TLH(CLK)}$ 9/	Fig. 6	126		GND	5.0 V	5.0 V	GND		GND	OUT	GND	5.0 V	5.0 V	IN_1	GND	5.0 V	CLK to Q		15		15		15	µs
$t_{TLH(CLK)}$ 9/		127						OUT									CLK to $\overline{Q}$							
$t_{THL(CLK)}$ 9/		128								OUT							CLK to Q							
$t_{THL(CLK)}$ 9/		129						OUT									CLK to $\overline{Q}$							
t_{SHL1}		130			IN_2					OUT							CLK to J_1		400		560		400	ns
		131			5.0 V	IN_2											CLK to J_2							
		132				5.0 V	IN_2										CLK to $\overline{J}_3$							
		133					GND				IN_2						CLK to $\overline{K}_3$							
		134									GND	IN_2					CLK to K_2							
		135										5.0 V	IN_2				CLK to K_1							
t_{SLH1}		136			IN_2								5.0 V				J_1 to CLK							
		137			5.0 V	IN_2											J_2 to CLK							
		138				5.0 V	IN_2										$\overline{J}_3$ to CLK							
		139					GND				IN_2						$\overline{K}_3$ to CLK							
		140					GND				GND	IN_2					K_2 to CLK							
		141					GND				GND	GND	IN_2				K_1 to CLK							
																		Subgroup 12 T_A = 25°C						
																		Min	Max					
t_{PHL1}	3003 Fig. 6	142		GND	10.0 V	10.0 V	GND		GND	OUT	GND	10.0 V	10.0 V	IN_1	GND	10.0 V	CLK to Q	10	200					ns
t_{PHL1}		143						OUT									CLK to $\overline{Q}$							
t_{PLH1}		144								OUT							CLK to Q							
t_{PLH1}		145						OUT									CLK to $\overline{Q}$							
t_{PHL2}		146		IN_1	GND	GND	10.0 V			OUT	10.0 V	GND	GND	GND	IN_2		S to Q	8	150					
t_{PHL2}		147		IN_1	GND	GND	10.0 V	OUT			10.0 V	GND	GND	GND	IN_2		S to $\overline{Q}$	8	150					

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TABLE III. Group A inspection for device type 03 -Continued.

Symbol	MIL-STD-883 method	Cases A,C,D,X,Y	Terminal conditions 1/														Measured terminal	Test limits						Units
		Symbol	NC	R	J_1	J_2	$\overline{J}_3$	$\overline{Q}$	V_{SS}	Q	$\overline{K}_3$	K_2	K_1	CLK	S	V_{DD}		Subgroup 12 T_A = 25°C						
		Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14		Min	Max	Min	Max	Min	Max	
t_{PLH2}	3003 Fig. 6	148		IN_1	GND	GND	10.0 V		GND	OUT	10.0 V	GND	GND	GND	GND	10.0 V	R to Q	8	150					ns
t_{PLH2}	3003 Fig. 6	149		IN_1	GND	GND	10.0 V	OUT			10.0 V	GND	GND	GND			R to $\overline{Q}$	8	150					
t_{THL}	3004 Fig. 6	150		GND	10.0 V	10.0 V	GND			OUT	GND	10.0 V	10.0 V	IN_1			CLK to Q	5	100					
t_{THL}		151						OUT									CLK to $\overline{Q}$							
t_{TLH}		152								OUT							CLK to Q							
t_{TLH}		153						OUT									CLK to $\overline{Q}$							

1/ Pins not designated may be "high" level logic, "low" level logic or open. Exceptions are as follows: $V_{IC(pos)}$ tests, the V_{SS} terminals shall be open; $V_{IC(neg)}$ tests, the V_{DD} terminal shall be open; I_{SS} tests, the output terminals shall be open.

2/ The I_{SS} measurements shall be performed in sequence.

3/ The device manufacturer may, at his option, measure I_{IL} and I_{IH} at 25°C for each individual input or measure all inputs together.

4/ The truth table tests shall be performed in sequence.

5/ The truth table tests shall be performed at V_{IH} and V_{DD} $\leq$5 Vdc and $\geq$18 Vdc. "L" = V_{SS} +0.5 V maximum and "H" = V_{DD} -0.5 V minimum.

6/ See 4.4.1c.

7/ The minimum clock frequency (f_{CLK}) requirement is considered met if proper output state changes occur with the pulse repetition period set to that given in the limits column.

8/ The minimum clock or reset or set pulse width ($t_{p(CLK)}$, $t_{p(R)}$, $t_{p(S)}$) requirement is considered met if proper output state changes occur with the pulse width set to that given in the limits column.

9/ Pulse repetition period = 100 µs, 50 percent duty cycle. The maximum clock transition time $t_{TLH(CLK)}$, $t_{THL(CLK)}$ requirement is considered met if proper output state changes occur with rise time set to that given in the limits column.

10/ Apply clock pulse; V_{IH} = 0 to 15 Vdc.

11/ Apply clock pulse; V_{IH} = 0 to 5 Vdc.

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12/ The input/output conditions and timing sequence shall apply:

Input/output conditions					
Test	V_{DD}	Input levels		Output levels	
		1	0	1	0
				Min	Max
V_{IL1} V_{IH1}	5.0 V	3.5 V	1.5 V	4.5 V	0.5 V
V_{IL2} V_{IH2}	10.0 V	7.0 V	3.0 V	9.0 V	1.0 V
V_{IL3} V_{IH3}	15.0 V	11.0 V	4.0 V	13.5 V	1.5 V

* Time slot		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36
INPUTS	R	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	0	0	0	0
	J_1	1	1	1	1	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0
	J_2	1	1	1	1	1	1	1	1	1	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0
	J_3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1
	K_3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1
	K_2	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	1	1	1	1	1	0	0	0	0
	K_1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1	0	0	0	0
	CLK	0	0	0	1	0	1	0	1	0	0	1	0	1	0	0	1	0	1	0	1	0	1	0	0	1	0	1	0	0	1	0	1	0	1	0	1
	S	1	0	0	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
OUTPUT	$\tilde{Q}$	0	1	1	0	0	1	1	1	0	0	1	1	1	0	0	1	1	1	1	0	0	0	1	1	0	0	0	1	1	0	0	0	0	1	0	1
	Q	1	0	0	1	1	0	0	0	1	1	0	0	0	1	1	0	0	0	0	1	1	1	0	0	1	1	1	0	0	1	1	1	1	1	1	1

* Tests are to be run in sequence.

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TABLE III. Group A inspection for device type 04.

Symbol	MIL-STD-883 method	Cases E,F,2	Terminal conditions 1/																Measured terminal	Test limits						Units
																				Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Symbol	CX_1	RC_1	R_1	$+TR_1$	$-TR_1$	Q_1	$\overline{Q}_1$	V_{SS}	$\overline{Q}_2$	Q_2	$-TR_2$	$+TR_2$	R_2	RC_2	CX_2	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16								
V_{IC} (pos)		1		1 mA														GND	RC_1		1.5					Vdc
		2			1 mA														R_1							
		3				1 mA													$+TR_1$							
		4					1 mA												$-TR_1$							
		5											1 mA						$-TR_2$							
		6												1 mA					$+TR_2$							
		7													1 mA				R_2							
		8														1 mA			RC_2							
V_{IC} (neg)		9		-1 mA						GND									RC_1		-6.0					
		10			-1 mA														R_1							
		11				-1 mA													$+TR_1$							
		12					-1 mA												$-TR_1$							
		13											-1 mA						$-TR_2$							
		14												-1 mA					$+TR_2$							
		15													-1 mA				R_2							
		16														-1 mA			RC_2							
I_{SS}	3005	17		8/	GND	GND	GND						GND	GND	GND	8/		18.0 V	V_{SS}		-.075		-2.5			µA
		18			18.0 V	GND	18.0 V						18.0 V	GND	18.0 V											
		19			18.0 V	18.0 V								18.0 V	18.0 V											
		20			GND	18.0 V								18.0 V	GND											
		21			18.0 V	GND								GND	18.0 V											
		22			18.0 V	GND	GND						GND	GND	18.0 V											
		23			GND	GND	GND						GND	GND	GND											

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TABLE III. Group A inspection for device type 04 - Continued.

Symbol	MIL-STD-883 method	Cases E,F,Z	Terminal conditions 1/															Measured terminal	Test limits						Units
																			Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Symbol	CX_1	RC_1	R_1	$+TR_1$	$-TR_1$	Q_1	$\overline{Q}_1$	V_{SS}	$\overline{Q}_2$	Q_2	$-TR_2$	$+TR_2$	R_2	RC_2	CX_2	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16								
V_{OH1}	3006	24		13.8 V	GND	GND	15.0 V			GND			15.0 V	GND	GND	13.8 V		15.0 V	$\overline{Q}_1$	14.95		14.95		14.95		Vdc
V_{OH1}	3006	25		7/	↓	↓	↓			↓			↓	↓	↓	7/		7/	$\overline{Q}_2$	14.95		14.95		14.95		
V_{OL1}	3007	26		↓	↓	↓	↓						↓	↓	↓	↓		↓	Q_1		0.05		0.05		0.05	
V_{OL1}	3007	27		↓	↓	↓	↓						↓	↓	↓	↓		↓	Q_2		0.05		0.05		0.05	
V_{OH1}	3006	28		GND	15.0 V	15.0 V								15.0 V	15.0 V	GND			Q_1	14.95		14.95		14.95		
V_{OH1}	3006	29		7/	↓	↓	↓							↓	↓	7/			Q_2	14.95		14.95		14.95		
V_{OL1}	3007	30		↓	↓	↓	↓						↓	↓	↓	↓		↓	$\overline{Q}_1$		0.05		0.05		0.05	
V_{OL1}	3007	31		↓	↓	↓	↓						↓	↓	↓	↓		↓	$\overline{Q}_2$		0.05		0.05		0.05	
V_{IL1}		32		7/	10/	10/	10/	10/	10/		10/	10/	10/	10/	10/	7/	10/	7/,10/	All outputs	10/	10/	10/	10/	10/	10/	
V_{IL2}		33		↓	↓	↓	↓	↓	↓		↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	
V_{IL3}		34		↓	↓	↓	↓	↓	↓		↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	
V_{IH1}		35		↓	↓	↓	↓	↓	↓		↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	
V_{IH2}		36		↓	↓	↓	↓	↓	↓		↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	
V_{IH3}		37		↓	↓	↓	↓	↓	↓		↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	
I_{OL1}		38		5.0 V	GND	GND	GND	0.4 V				0.4 V	GND	GND	GND	5.0 V		5.0 V	Q_1	0.51		0.36		0.64		mA
		39		5.0 V	GND	↓	GND	↓				↓	GND	↓	GND	5.0 V		↓	Q_2	↓		↓		↓		↓
		40		8/	GND	↓	5.0 V	↓				↓	5.0 V	↓	GND	8/		↓	None	↓		↓		↓		↓
		41		↓	5.0 V	↓	↓	↓				↓	↓	↓	5.0 V	↓		↓	None	↓		↓		↓		↓
		42		↓	5.0 V	5.0 V	↓	↓		0.4 V		↓	↓	5.0 V	5.0 V	↓		↓	$\overline{Q}_1$	↓		↓		↓		↓
		43		↓	5.0 V	5.0 V	↓	↓		↓	0.4 V	↓	↓	5.0 V	5.0 V	↓		↓	$\overline{Q}_2$	↓		↓		↓		↓
I_{OH1}		44		4.6 V	GND	GND	GND						GND	GND	GND	5.0 V			RC_1	-0.51		-0.36		-0.64		
		45		5.0 V	↓	↓	↓		4.6 V				↓	↓	↓	5.0 V			$\overline{Q}_1$	↓		↓		↓		
		46		5.0 V	↓	↓	↓		↓		4.6 V		↓	↓	↓	5.0 V			$\overline{Q}_2$	↓		↓		↓		
		47		5.0 V	↓	↓	↓		↓	↓	↓		↓	↓	↓	4.6 V			RC_2	↓		↓		↓		
		48		8/	↓	↓	5.0 V		↓	↓	↓		5.0 V	↓	↓	8/		↓	None	↓		↓		↓		↓

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TABLE III. Group A inspection for device type 04 - Continued.

Symbol	MIL-STD-883 method	Cases E,F,Z	Terminal conditions 1/															Measured terminal	Test limits						Units
																			Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Symbol	CX_1	RC_1	R_1	$+TR_1$	$-TR_1$	Q_1	$\overline{Q}_1$	V_{SS}	$\overline{Q}_2$	Q_2	$-TR_2$	$+TR_2$	R_2	RC_2	CX_2	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16								
I_{OH1}		49		8/	5.0 V	GND	5.0 V			GND			5.0 V	GND	5.0 V	8/		5.0 V	None	-0.51		-0.36		-0.64		mA
		50		8/	5.0 V	5.0 V	5.0 V	4.6 V					5.0 V	5.0 V		8/		5.0 V	Q_1							
		51		8/	5.0 V	5.0 V	5.0 V					4.6 V	5.0 V	5.0 V		8/		5.0 V	Q_2							
I_{OL2}		52		15.0 V	GND	GND	GND	1.5 V					GND	GND	GND	15.0 V		15.0 V	Q_1	3.4		2.4		4.2		
		53		15.0 V	GND		GND					1.5 V	GND			15.0 V			Q_2							
		54		8/	GND		15.0 V						15.0 V			8/			None							
		55			15.0 V										15.0 V				None							
		56			15.0 V	15.0 V			1.5 V					15.0 V	15.0 V				$\overline{Q}_1$							
		57			15.0 V	15.0 V					1.5 V			15.0 V	15.0 V				$\overline{Q}_2$							
I_{OH2}		58		13.5 V	GND	GND	GND						GND	GND	GND	15.0 V			RC_1	-3.4		-2.4		-4.2		
		59		15.0 V					13.5 V							15.0 V			$\overline{Q}_1$							
		60		15.0 V							13.5 V					15.0 V			$\overline{Q}_2$							
		61		15.0 V												13.5 V			RC_2							
		62		8/			15.0 V						15.0 V			8/			None							
		63			15.0 V										15.0 V				None							
		64			15.0 V	15.0 V		13.5 V						15.0 V	15.0 V				Q_1							
		65			15.0 V	15.0 V						13.5 V		15.0 V	15.0 V				Q_2							
I_{IH1} 3/	3010	66			18.0 V	18.0 V	18.0 V						18.0 V	18.0 V	18.0 V			18.0 V	All inputs together		6					nA
I_{IH2}		67			18.0 V	GND	GND						GND	GND	GND				R_1		t		45			
		68			GND	18.0 V	GND						GND						$+TR_1$							
		69				GND	18.0 V						GND						$-TR_1$							
		70					GND						18.0 V						$-TR_2$							
		71											GND	18.0 V					$+TR_2$							
		72											GND	GND	18.0 V				R_2							
I_{IL1} 3/	3009	73											GND	GND	GND				All inputs together		-6					

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TABLE III. Group A inspection for device type 04 - Continued.

Symbol	MIL-STD-883 method	Cases E,F,Z	Terminal conditions 1/																Measured terminal	Test limits						Units
																				Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Symbol	CX_1	RC_1	R_1	$+TR_1$	$-TR_1$	Q_1	$\overline{Q}_1$	V_{SS}	$\overline{Q}_2$	Q_2	$-TR_2$	$+TR_2$	R_2	RC_2	CX_2	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16								
I_{IL2}	3009	74			GND	18.0 V	18.0 V			GND			18.0 V	18.0 V	18.0 V			18.0 V	R_1		-1		-45			nA
		75			18.0 V	GND	18.0 V						18.0 V						$+TR_1$							
		76				18.0 V	GND						18.0 V						$-TR_1$							
		77					18.0 V						GND						$-TR_2$							
		78					18.0 V						18.0 V	GND					$+TR_2$							
		79					18.0 V						18.0 V	18.0 V	GND				R_2							
																				Subgroup 4 T_A = 25°C						
																				Min	Max					
C_I	3012	80			6/					GND								GND	R_1		7.5					pF
		81				6/													$+TR_1$							
		82					6/												$-TR_1$							
		83											6/						$-TR_2$							
		84												6/					$+TR_2$							
		85													6/				R_2							
																				Subgroup 7 T_A = 25°C		Subgroup 8 T_A = 125°C		Subgroup 8 T_A = -55°C		
																				Min	Max	Min	Max	Min	Max	
Truth table tests	3014	86		5.0 V	GND	GND	5.0 V	L	H	GND	H	L	5.0 V	GND	GND	5.0 V		5/	All outputs	See notes 4/, 5/, 7/						
		87		GND 7/	5.0 V		5.0 V	H	L		L	H	5.0 V		5.0 V	GND										
		88		5.0 V	GND		GND	L	H		H	L	GND		GND	5.0 V										
		89					5.0 V	L	H		H	L	5.0 V													
		90				5.0 V		L	H		H	L		5.0 V												
		91			5.0 V	5.0 V		L	H		H	L		5.0 V	5.0 V											
		92			5.0 V	GND		H	L		L	H		GND	5.0 V											
		93			GND			L	H		H	L			GND											
		94			5.0 V			L	H		H	L			5.0 V											
		95			5.0 V		GND	H	L		L	H	GND		5.0 V											
		96			GND			L	H		H	L			GND											
		97			5.0 V			L	H		H	L			5.0 V											
		98			5.0 V		5.0 V	H	L		L	H	5.0 V		5.0 V											
		99			GND			L	H		H	L			GND											
		100			5.0 V			L	H		H	L			5.0 V											
		101			5.0 V	5.0 V		H	L		L	H		5.0 V	5.0 V											

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TABLE III. Group A inspection for device type 04 - Continued.

Symbol	MIL-STD-883 method	Cases E,F,Z	Terminal conditions 1/																Measured terminal	Test limits						Units
																				Subgroup 9 T_A = 25°C		Subgroup 10 T_A = 125°C		Subgroup 11 T_A = -55°C		
		Symbol	CX_1	RC_1	R_1	$+TR_1$	$-TR_1$	Q_1	$\overline{Q}_1$	V_{SS}	$\overline{Q}_2$	Q_2	$-TR_2$	$+TR_2$	R_2	RC_2	CX_2	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16								
t_{PLH1}	3003 Fig. 6	102		9/	5.0 V	IN	5.0 V	OUT		GND			5.0 V	GND	5.0 V	9/		5.0 V	$+TR_1$ to Q_1	25	500	35	700	25	500	ns
		103				GND	5.0 V					OUT	5.0 V	IN					$+TR_2$ to Q_2							
		104				GND	IN	OUT					5.0 V	GND					$-TR_1$ to Q_1							
		105				GND	5.0 V					OUT	IN	GND					$-TR_2$ to Q_2							
t_{PHL1}		106				IN	5.0 V		OUT				5.0 V	GND					$+TR_1$ to $\overline{Q}_1$							
		107				GND	5.0 V				OUT		5.0 V	IN					$+TR_2$ to $\overline{Q}_2$							
		108				GND	IN		OUT				5.0 V	GND					$-TR_1$ to $\overline{Q}_1$							
		109				GND	5.0 V				OUT		IN	GND					$-TR_2$ to $\overline{Q}_2$							
t_{PHL2}		110			IN	IN	5.0 V	OUT					5.0 V	IN	IN				R_1 to Q_1	23	450	32	630	23	450	
t_{PHL2}		111										OUT							R_2 to Q_2							
t_{PLH2}		112							OUT										R_1 to $\overline{Q}_1$							
t_{PLH2}		113									OUT								R_2 to $\overline{Q}_2$							
t_{TLH1}	3004 Fig. 6	114			5.0 V			OUT							5.0 V				Q_1	10	200	14	280	10	200	
t_{TLH1}		115										OUT							Q_2							
t_{THL1}		116							OUT										$\overline{Q}_1$							
t_{THL1}		117									OUT								$\overline{Q}_2$							
t_{THL2}		118			IN			OUT							IN				Q_1							
		119						OUT											Q_1	15	300	21	420	15	300	
		120						OUT											Q_1	25	500	35	700	25	500	
		121										OUT							Q_2	10	200	14	280	10	200	
		122										OUT							Q_2	15	300	21	420	15	300	

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TABLE III. Group A inspection for device type 04 - Continued.

Symbol	MIL-STD-883 method	Cases E,F,Z	Terminal conditions 1/															Measured terminal	Test limits						Units
																			Subgroup 9 T_A = 25°C		Subgroup 10 T_A = 125°C		Subgroup 11 T_A = -55°C		
		Symbol	CX_1	RC_1	R_1	$+TR_1$	$-TR_1$	Q_1	$\overline{Q}_1$	V_{SS}	$\overline{Q}_2$	Q_2	$-TR_2$	$+TR_2$	R_2	RC_2	CX_2	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16								
t_{THL2}	3004 Fig. 6	123		9/	IN	IN	5.0 V			GND		OUT	5.0 V	IN	IN	9/		5.0 V	Q_2	25	500	35	700	25	700	ns
t_{TLH2}		124			5.0 V				OUT						5.0 V				$\overline{Q}_1$	10	200	14	280	10	200	
		125							OUT										$\overline{Q}_1$	15	300	21	420	15	300	
		126							OUT										$\overline{Q}_1$	25	500	35	700	25	500	
		127									OUT								$\overline{Q}_2$	10	200	14	280	10	200	
		128									OUT								$\overline{Q}_2$	15	300	21	420	15	300	
		129									OUT								$\overline{Q}_2$	25	500	35	700	25	500	
																				Subgroup 12 T_A = 25°C						
																				Min	Max					
t_{PLH1}	3003 Fig. 6	130		9/	10.0 V	IN	10.0 V	OUT		GND			10.0 V	GND	10.0 V	9/		10.0 V	$+TR_1$ to Q_1	13	250					ns
		131				GND	10.0 V					OUT	10.0 V	IN					$+TR_2$ to Q_2							
		132				GND	IN	OUT					10.0 V	GND					$-TR_1$ to Q_1							
		133				GND	10.0 V					OUT	IN	GND					$-TR_2$ to Q_2							
t_{PHL1}		134				IN	10.0 V						10.0 V	GND					$+TR_1$ to $\overline{Q}_1$							
		135				GND	10.0 V						10.0 V	IN					$+TR_2$ to $\overline{Q}_2$							
		136				GND	IN						10.0 V	GND					$-TR_1$ to $\overline{Q}_1$							
		137				GND	10.0 V						IN	GND					$-TR_2$ to $\overline{Q}_2$							
t_{PHL2}		138			IN	IN	10.0 V	OUT					10.0 V	IN	IN				R_1 to Q_1							
t_{PHL2}		139										OUT							R_2 to Q_2							
t_{PLH2}		140							OUT										R_1 to $\overline{Q}_1$							
t_{PLH2}		141									OUT								R_2 to $\overline{Q}_2$							
t_{TLH1}	3004 Fig. 6	142			10.0 V			OUT							10.0 V				Q_1	5	100					μs
t_{TLH1}		143										OUT							Q_2							
t_{THL1}		144							OUT										$\overline{Q}_1$							
t_{THL1}		145									OUT								$\overline{Q}_2$							

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TABLE III. Group A inspection for device type 04 - Continued.

Symbol	MIL-STD-883 method	Cases E,F,Z	Terminal conditions 1/																Measured terminal	Test limits						Units
																				Subgroup 12 T_A = 25°C						
		Symbol	CX_1	RC_1	R_1	$+TR_1$	$-TR_1$	Q_1	$\overline{Q}_1$	V_{SS}	$\overline{Q}_2$	Q_2	$-TR_2$	$+TR_2$	R_2	RC_2	CX_2	V_{DD}								
		Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Min	Max	Min	Max	Min	Max	
t_{THL2}	3004 Fig. 6	146		9/	IN	IN	10.0 V	OUT		GND			10.0 V	IN	IN	9/		10.0 V	Q_1	5	100					ns
		147						OUT											Q_1	8	150					
		148						OUT											Q_1	15	300					
		149										OUT							Q_2	5	100					
		150										OUT							Q_2	8	150					
		151										OUT							Q_2	15	300					
t_{TLH2}		152			10.0 V				OUT						10.0 V				$\overline{Q}_1$	5	100					
		153							OUT										$\overline{Q}_1$	8	150					
		154							OUT										$\overline{Q}_1$	15	300					
		155									OUT								$\overline{Q}_2$	5	100					
		156									OUT								$\overline{Q}_2$	8	150					
		157									OUT								$\overline{Q}_2$	15	300					

1/ Pins not designated may be "High" level logic, "Low" level logic or open. Exceptions are as follows: $V_{IC(pos)}$ tests, the V_{SS} terminals shall be open; $V_{IC(neg)}$ tests, the V_{DD} terminal shall be open; I_{SS} tests, the output terminals shall be open.

2/ The I_{SS} measurements shall be performed in sequence.

3/ The device manufacturer may, at his option, measure I_{IL} and I_{IH} at 25°C for each individual input or measure all inputs together.

4/ The truth table tests shall be performed in sequence.

5/ The truth table tests shall be performed at V_{IH} and $V_{DD} \leq 5$ Vdc and ≥ 18 Vdc. "L" = V_{SS} + 0.5 V maximum and "H" = V_{DD} -0.5 V minimum.

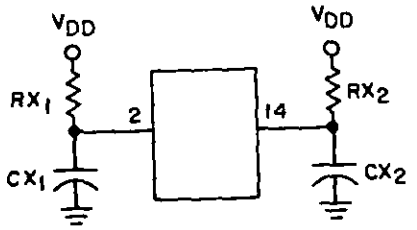
6/ See 4.4.1c.

7/ 100 kΩ ±10 percent resistor shall be connected from V_{DD} to pin 2 and another one connected from V_{DD} to pin 14, similarly 0.01 µF ±20 percent capacitors shall be connected from pins 1 and 2 and between pins 14 and 15 to provide timing for all V_O, V_I, and truth table tests. Optionally, pins 2 and 14 may be GND or 13.8 V for the V_{OH1} and V_{OL1} tests, and GND or V_{DD} for the truth table tests, per table III.

8/ A 100 kΩ ±10 percent resistor shall be connected from pin 2 to GND and another one connected from pin 14 to GND.

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9/ Connect RX_1, CX_1, RX_2, and CX_2 using the values in the table below:



Test	$RX_{1,2}$	$CX_{1,2}$
	KΩ ±10%	pF ±10%
t_{PLH1}	10	100
t_{PHL1}	10	100
t_{PLH2} *	10	100
t_{PLH2} *	10	100
t_{TLH1}	10	100
t_{THL1}	10	100
t_{THL2}	10	0.01 µF
t_{THL2}	10	0.1 µF
t_{THL2}	10	1.0 µF
t_{TLH2}	10	0.01 µF
t_{TLH2}	10	0.1 µF
t_{TLH2}	10	1.0 µF

* Reset pulse shall be applied before the RXCX times out.

10/ The following timing sequence and input/output conditions shall apply:

Input/output conditions

Test	V_{DD}	Input levels		Output levels		RX_1, RX_2 Input levels	
		1	0	1	0	1	0
V_{IL1} V_{IH1}	5.0 V	Max 3.5 V	Min 1.5 V	Min 4.5 V	Max 0.5 V	V_{DD}	V_{SS}
V_{IL2} V_{IH2}	10.0 V	7.0 V	3.0 V	9.0 V	1.0 V		
V_{IL3} V_{IH3}	15.0 V	11.0 V	4.0 V	13.5 V	1.5 V		

	* Time slot	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
INPUTS	RC_1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1
	R_1	0	1	0	0	0	1	1	0	1	1	0	1	1	0	1	1
	$+TR_1$	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	1
	$-TR_1$	1	1	0	1	1	1	1	1	1	0	0	0	1	1	1	1
	$-TR_2$	1	1	0	1	1	1	1	1	1	0	0	0	1	1	1	1
	$+TR_2$	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	1
	R_2	0	1	0	0	0	1	1	0	1	1	0	1	1	0	1	1
	RC_2	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1
OUTPUTS	Q_1	0	1	0	0	0	0	1	0	0	1	0	0	1	0	0	1
	$\overline{Q}_1$	1	0	1	1	1	1	0	1	1	0	1	1	0	1	1	0
	Q_2	0	1	0	0	0	0	1	0	0	1	0	0	1	0	0	1
	$\overline{Q}_2$	1	0	1	1	1	1	0	1	1	0	1	1	0	1	1	0

* Tests shall be run in sequence.

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TABLE III. Group A inspection for device type 05.

Symbol	MIL-STD-883 method	Cases E,F,Z	Terminal conditions 1/															Measured terminal	Test limits						Units
																			Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Symbol	$\overline{C}$	Q_1	D_1	D_2	Q_2	D_3	Q_3	V_{SS}	CLK	Q_4	D_4	Q_5	D_5	D_6	Q_6	V_{DD}							
		Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Min	Max	Min	Max	Min	Max	
V_{IC} (pos)		1	1 mA															GND	$\overline{C}$		1.5					Vdc
		2			1 mA														D_1							
		3				1 mA													D_2							
		4						1 mA											D_3							
		5									1 mA								CLK							
		6											1 mA						D_4							
		7													1 mA				D_5							
		8														1 mA			D_6							
V_{IC} (neg)		9	-1 mA							GND									$\overline{C}$		-6.0					
		10			-1 mA														D_1							
		11				-1 mA													D_2							
		12						-1 mA											D_3							
		13									-1 mA								CLK							
		14											-1 mA						D_4							
		15													-1 mA				D_5							
		16														-1 mA			D_6							
I_{SS}	3005	17	GND		18.0 V	18.0 V		18.0 V			GND		18.0 V		18.0 V	18.0 V		18.0 V	V_{SS}		-.075		-2.5			µA
		18	18.0 V		18.0 V	18.0 V		18.0 V			7/		18.0 V		18.0 V	18.0 V										
		19	18.0 V		GND	GND		GND			7/		GND		GND	GND										
		20	18.0 V		GND	GND		GND			GND		GND		GND	GND										
V_{OH1}	3006	21	15.0 V		15.0 V	15.0 V		15.0 V			8/ 15.0 V		15.0 V		15.0 V	15.0 V		15.0 V	Q_1	14.95		14.95		14.95		Vdc
		22																	Q_2							
		23																	Q_3							
		24																	Q_4							
		25																	Q_5							
		26																	Q_6							

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TABLE III. Group A inspection for device type 05 -Continued.

Symbol	MIL-STD-883 method	Cases E,F,Z	Terminal conditions 1/															Measured terminal	Test limits						Units
																			Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Symbol	$\overline{C}$	Q_1	D_1	D_2	Q_2	D_3	Q_3	V_{SS}	CLK	Q_4	D_4	Q_5	D_5	D_6	Q_6	V_{DD}							
		Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Min	Max	Min	Max	Min	Max	
V_{OL1}	3007	27	15.0 V		GND	GND		GND		GND	8/ 15.0 V		GND		GND	GND		15.0 V	Q_1		0.05		0.05		0.05	Vdc
		28																	Q_2							
		29																	Q_3							
		30																	Q_4							
		31																	Q_5							
		32																	Q_6							
		33	GND		15.0 V	15.0 V		15.0 V			8/ 15.0 V		15.0 V		15.0 V	15.0 V			Q_1							
		34																	Q_2							
		35																	Q_3							
		36																	Q_4							
		37																	Q_5							
		38																	Q_6							
V_{IL1}		39	13/	13/	13/	13/	13/	13/	13/		13/	13/	13/	13/	13/	13/	13/	13/	All outputs	13/	13/	13/	13/	13/	13/	
V_{IL2}		40																								
V_{IL3}		41																								
V_{IH1}		42																								
V_{IH2}		43																								
V_{IH3}		44																								
I_{OL1}		45	5.0 V	0.4 V	GND	GND		GND			9/ 5.0 V		GND		GND	GND		5.0 V	Q_1	0.51		0.36		0.64		mA
		46					0.4 V												Q_2							
		47							0.4 V			0.4 V							Q_3							
		48												0.4 V					Q_4							
		49															0.4 V		Q_5							
		50																	Q_6							
I_{OH1}		51		4.6 V	5.0 V	5.0 V		5.0 V			9/		5.0 V		5.0 V	5.0 V			Q_1	-0.51		-0.36		-0.64		

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TABLE III. Group A inspection for device type 05 -Continued.

Symbol	MIL-STD-883 method	Cases E,F,2	Terminal conditions 1/															Measured terminal	Test limits						Units
																			Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Symbol	$\overline{C}$	Q_1	D_1	D_2	Q_2	D_3	Q_3	V_{SS}	CLK	Q_4	D_4	Q_5	D_5	D_6	Q_6	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16								
I_{OH1}		52	5.0 V		5.0 V	5.0 V	4.6 V	5.0 V		GND	5.0 V		5.0 V		5.0 V	5.0 V		5.0 V	Q_2	-0.51		-0.36		-0.64		mA
		53							4.6 V										Q_3							
		54										4.6 V							Q_4							
		55												4.6 V					Q_5							
		56															4.6 V		Q_6							
I_{OL2}		57	15.0 V	1.5 V	GND	GND		GND			8/		GND		GND	GND		15.0 V	Q_1	3.4		2.4		4.2		
		58					1.5 V				15.0 V								Q_2							
		59							1.5 V										Q_3							
		60										1.5 V							Q_4							
		61												1.5 V					Q_5							
		62															1.5 V		Q_6							
I_{OH2}		63		13.5 V	15.0 V	15.0 V		15.0 V			8/		15.0 V		15.0 V	15.0 V			Q_1	-3.4		-2.4		-4.2		
		64					13.5 V				15.0 V								Q_2							
		65							13.5 V										Q_3							
		66										13.5 V							Q_4							
		67												13.5 V					Q_5							
		68															13.5 V		Q_6							
I_{IH1} 3/	3010	69	18.0 V		18.0 V	18.0 V		18.0 V			18.0 V		18.0 V		18.0 V	18.0 V		18.0 V	All inputs together		8				nA
I_{IH2}		70	18.0 V		GND	GND		GND			GND		GND		GND	GND			$\overline{C}$		1		45		
		71	GND		18.0 V	GND		GND											D_1							
		72			GND	18.0 V		GND											D_2							
		73				GND		18.0 V											D_3							
		74						GND			18.0 V								CLK							
		75									GND		18.0 V						D_4							
		76											GND		18.0 V				D_5							
		77											GND		GND	18.0 V			D_6							

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TABLE III. Group A inspection for device type 05 -Continued.

Symbol	MIL-STD-883 method	Cases E,F,Z	Terminal conditions 1/															Measured terminal	Test limits						Units
																			Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Symbol	$\overline{C}$	Q_1	D_1	D_2	Q_2	D_3	Q_3	V_{SS}	CLK	Q_4	D_4	Q_5	D_5	D_6	Q_6	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16								
I_{IL1} 3/	3009	78	GND		GND	GND		GND		GND	GND		GND		GND	GND		18.0 V	All inputs together		-8					nA
I_{IL2}		79	GND		18.0 V	18.0 V		18.0 V			18.0 V		18.0 V		18.0 V	18.0 V			$\overline{C}$		-1		-45			
		80	18.0 V		GND			18.0 V											D_1							
		81			18.0 V	GND		18.0 V											D_2							
		82				18.0 V		GND											D_3							
		83						18.0 V			GND								CLK							
		84									18.0 V		GND						D_4							
		85									18.0 V		18.0 V		GND				D_5							
		86									18.0 V		18.0 V		18.0 V	GND			D_6							
																				Subgroup 4 T_A = 25°C						
																				Min	Max					
C_I	3012	87	6/		6/					GND								GND	$\overline{C}$		15					pF
		88				6/													D_1		7.5					
		89						6/											D_2							
		90									6/								D_3							
		91											6/						CLK							
		92													6/				D_4							
		93														6/			D_5							
		94																	D_6							
																				Subgroup 7 T_A = 25°C		Subgroup 8 T_A = 25°C		Subgroup 8 T_A = -55°C		
																				Min	Max	Min	Max	Min	Max	
Truth table test	3014	95	GND	L	5.0 V	5.0 V	L	5.0 V	L	GND	GND	L	5.0 V	L	5.0 V	5.0 V	L	5.0 V	All outputs	See notes 4/, 5/						
		96	5.0 V	L			L		L		GND	L		L			L									
		97	5.0 V	H			H		H		5.0 V	H		H			H									
		98	5.0 V	H			H		H		GND	H		H			H									
		99	GND	L			L		L		GND	L		L			L									

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TABLE III. Group A inspection for device type 05 -Continued.

Symbol	MIL-STD-883 method	Cases E,F,Z	Terminal conditions 1/																Measured terminal	Test limits						Units
																				Subgroup 7 T_A = 25°C		Subgroup 8 T_A = 125°C		T_A = -55°C		
		Symbol	$\overline{C}$	Q_1	D_1	D_2	Q_2	D_3	Q_3	V_{SS}	CLK	Q_4	D_4	Q_5	D_5	D_6	Q_6	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16								
Truth table test	3014	100	GND	L	5.0 V	5.0 V	L	5.0 V	L	GND	5.0 V	L	5.0 V	L	5.0 V	5.0 V	L	5.0 V	All outputs	See notes 4/, 5/						
		101	5.0 V	L	5.0 V	5.0 V	L	5.0 V	L		GND	L	5.0 V	L	5.0 V	5.0 V	L									
		102		H	5.0 V	5.0 V	H	5.0 V	H		5.0 V	H	5.0 V	H	5.0 V	5.0 V	H									
		103		H	GND	GND	H	GND	H		GND	H	GND	H	GND	GND	H									
		104		L	GND	GND	L	GND	L		5.0 V	L	GND	L	GND	GND	L									
		105		L	5.0 V	5.0 V	L	5.0 V	L		GND	L	5.0 V	L	5.0 V	5.0 V	L									
		106		H	5.0 V	5.0 V	H	5.0 V	H		5.0 V	H	5.0 V	H	5.0 V	5.0 V	H									
		107	GND	L	5.0 V	5.0 V	L	5.0 V	L		5.0 V	L	5.0 V	L	5.0 V	5.0 V	L									
		108	GND	L	GND	GND	L	GND	L		GND	L	GND	L	GND	GND	L									
																				Subgroup 9 T_A = 25°C		Subgroup 10 T_A = 125°C		Subgroup 11 T_A = -55°C		
																				Min	Max	Min	Max	Min	Max	
t_{PHL1}	3003 Fig. 6	109	5.0 V	OUT	IN_2	GND	OUT	GND	OUT	GND	IN_1	OUT	GND	OUT	GND	GND		5.0 V	CLK to Q_1	15	300	21	420	15	300	ns
		110			GND	IN_2		GND					GND						CLK to Q_2							
		111				GND		IN_2					GND						CLK to Q_3							
		112						GND					IN_2						CLK to Q_4							
		113											GND		GND				CLK to Q_5							
		114													IN_2	IN_2	OUT		CLK to Q_6							
t_{PHL2}		115	IN_2	OUT	5.0 V										GND				$\overline{C}$ to Q_1	15	300	21	420	15	300	
		116			GND	5.0 V	OUT												$\overline{C}$ to Q_2							
		117				GND		5.0 V	OUT										$\overline{C}$ to Q_3							
		118						GND				OUT	5.0 V						$\overline{C}$ to Q_4							
		119											GND	OUT	5.0 V				$\overline{C}$ to Q_5							
		120													GND	5.0 V	OUT		$\overline{C}$ to Q_6							
t_{PLH1}		121	5.0 V	OUT	IN_2						IN_1					GND			CLK to Q_1	15	300	21	420	15	300	
		122	5.0 V		GND	IN_2	OUT				IN_1					GND			CLK to Q_2	15	300	21	420	15	300	
		123	5.0 V		GND	GND		IN_2	OUT		IN_1					GND			CLK to Q_3	15	300	21	420	15	300	

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TABLE III. Group A inspection for device type 05 -Continued.

Symbol	MIL-STD-883 method	Cases E,F,Z	Terminal conditions 1/																Measured terminal	Test limits						Units
																				Subgroup 9 T_A = 25°C		Subgroup 10 T_A = 125°C		Subgroup 11 T_A = -55°C		
		Symbol	$\overline{E}$	Q_1	D_1	D_2	Q_2	D_3	Q_3	V_{SS}	CLK	Q_4	D_4	Q_5	D_5	D_6	Q_6	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16								
t_{PLH1}	3003 Fig. 6	124	5.0 V		GND	GND		GND		GND	IN_1	OUT	IN_2	OUT	GND	GND		5.0 V	CLK to Q_4	15	300	21	420	15	300	ns
		125	5.0 V		GND								GND		IN_2	GND			CLK to Q_5	15	300	21	420	15	300	
		126	5.0 V		GND										GND	IN_2	OUT		CLK to Q_6	15	300	21	420	15	300	
t_{THL}	3004 Fig. 6	127	5.0 V	OUT	IN_2	IN_2	OUT	IN_2	OUT			OUT	IN_2	OUT	IN_2	GND			Q_1	10	200	14	280	10	200	
		128			GND	GND		GND					GND		GND				Q_2							
		129																	Q_3							
		130																	Q_4							
		131																	Q_5							
		132														IN_2	OUT		Q_6							
t_{TLH}		133		OUT	IN_2	IN_2	OUT	IN_2	OUT			OUT	IN_2	OUT	IN_2	GND			Q_1							
		134			GND	GND		GND					GND		GND				Q_2							
		135																	Q_3							
		136																	Q_4							
		137																	Q_5							
		138														IN_2	OUT		Q_6							
f_{CLK} 10/	Fig. 6	139		OUT	IN_2	IN_2	OUT	IN_2	OUT			OUT	IN_2	OUT	IN_2	GND			CLK to Q_1		570		800		570	
		140			GND	GND		GND					GND		GND				CLK to Q_2							
		141																	CLK to Q_3							
		142																	CLK to Q_4							
		143																	CLK to Q_5							
		144														IN_2	OUT		CLK to Q_6							

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TABLE III. Group A inspection for device type 05 -Continued.

Symbol	MIL-STD-883 method	Cases E,F,Z	Terminal conditions 1/																Measured terminal	Test limits						Units
																				Subgroup 9 T_A = 25°C		Subgroup 10 T_A = 125°C		Subgroup 11 T_A = -55°C		
		Symbol	$\overline{C}$	Q_1	D_1	D_2	Q_2	D_3	Q_3	V_{SS}	CLK	Q_4	D_4	Q_5	D_5	D_6	Q_6	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16								
t_p(CLK) 11/	Fig. 6	145	5.0 V	OUT	IN_2	GND	OUT	GND	OUT	GND	IN_1	OUT	GND	OUT	GND	GND	OUT	5.0 V	CLK to Q_1		200		280		200	ns
		146			GND	IN_2		GND					GND						CLK to Q_2							
		147				GND		IN_2					GND						CLK to Q_3							
		148						GND					IN_2						CLK to Q_4							
		149						GND					GND		IN_2				CLK to Q_5							
		150						GND					GND		GND	IN_2	OUT		CLK to Q_6							
t_p(CLR) 11/		151	IN_2	OUT	5.0 V	5.0 V	OUT	5.0 V	OUT			OUT	5.0 V	OUT	5.0 V	5.0 V	OUT		$\overline{C}$ to Q_1							
		152																	$\overline{C}$ to Q_2							
		153																	$\overline{C}$ to Q_3							
		154																	$\overline{C}$ to Q_4							
		155																	$\overline{C}$ to Q_5							
		156																	1 to 15							
t_{SLH}		157	5.0 V		IN_2	GND		GND					GND		GND	GND			D_1 to CLK		80		120		80	
		158			GND	IN_2		GND					GND						D_2 to CLK							
		159				GND		IN_2					GND						D_3 to CLK							
		160						GND					IN_2						D_4 to CLK							
		161											GND		IN_2				D_5 to CLK							
		162													GND	IN_2			D_6 to CLK							
t_{SHL}		163			IN_2											GND			CLK to D_1							
		164			GND	IN_2													CLK to D_2							
		165				GND		IN_2											CLK to D_3							
		166						GND					IN_2						CLK to D_4							
		167											GND		IN_2				CLK to D_5							
		168													GND	IN_2			CLK to D_6							
t_{HHL}		169			IN_2											GND			CLK to D_1		60		85		60	
t_{HHL}		170			GND	IN_2										GND			CLK to D_2		60		85		60	
t_{HHL}		171			GND	GND		IN_2								GND			CLK to D_3		60		85		60	

See footnotes at end of device type 05.

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TABLE III. Group A inspection for device type 05 -Continued.

Symbol	MIL-STD-883 method	Cases E,F,Z	Terminal conditions 1/																Measured terminal	Test limits						Units
																				Subgroup 9 T_A = 25°C		Subgroup 10 T_A = 125°C		Subgroup 11 T_A = -55°C		
		Symbol	$\overline{C}$	Q_1	D_1	D_2	Q_2	D_3	Q_3	V_{SS}	CLK	Q_4	D_4	Q_5	D_5	D_6	Q_6	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16								
t_{HHL} t_{HHL} t_{HHL}	Fig. 6	172 173 174	5.0 V		GND	GND		GND		GND	IN_1		IN_2 GND		GND IN_2 GND	GND GND IN_2		5.0 V	CLK to D_4 CLK to D_5 CLK to D_6		60		85		60	ns
t_{HLH}		175 176 177 178 179 180			IN_2 GND	IN_2 GND		IN_2 GND					IN_2 GND		IN_2 GND	GND IN_2			D_1 to CLK D_2 to CLK D_3 to CLK D_4 to CLK D_5 to CLK D_6 to CLK							
$t_{TLH(CLK)}$ 12/		181 182 183 184 185 186		OUT	IN_2 GND	IN_2 GND	OUT	IN_2 GND	OUT			OUT	IN_2 GND	OUT	IN_2 GND	GND IN_2	OUT		CLK to Q_1 CLK to Q_2 CLK to Q_3 CLK to Q_4 CLK to Q_5 CLK to Q_6		15		15		15	µs
$t_{THL(CLK)}$ 12/		187 188 189 190 191 192		OUT	IN_2 GND	IN_2 GND	OUT	IN_2 GND GND GND	OUT			OUT	IN_2 GND GND	OUT	IN_2 GND	GND IN_2	OUT		CLK to Q_1 CLK to Q_2 CLK to Q_3 CLK to Q_4 CLK to Q_5 CLK to Q_6							

See footnotes at end of device type 05.

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TABLE III. Group A inspection for device type 05 -Continued.

Symbol	MIL-STD-883 method	Cases E,F,Z	Terminal conditions 1/																Measured terminal	Test limits						Units
																				Subgroup 12 T_A = 25°C						
		Symbol	$\overline{C}$	Q_1	D_1	D_2	Q_2	D_3	Q_3	V_{SS}	CLK	Q_4	D_4	Q_5	D_5	D_6	Q_6	V_{DD}								
		Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Min	Max	Min	Max	Min	Max	
t_{PHL1}	3003 Fig. 6	193	10.0 V	OUT	IN_2	GND		GND		GND	IN_1		GND		GND	GND		10.0 V	CLK to Q_1	5	110					ns
		194			GND	IN_2	OUT	GND					GND						CLK to Q_2							
		195				GND		IN_2	OUT				GND						CLK to Q_3							
		196						GND				OUT	IN_2						CLK to Q_4							
		197											GND	OUT	IN_2	IN_2	OUT		CLK to Q_5							
		198													GND	GND			CLK to Q_6							
t_{PHL2}		199	IN_2	OUT	10.0 V											GND			$\overline{C}$ to Q_1							
		200			GND	10.0 V	OUT												$\overline{C}$ to Q_2							
		201				GND		10.0 V	OUT										$\overline{C}$ to Q_3							
		202						GND				OUT	10.0 V						$\overline{C}$ to Q_4							
		203											GND	OUT	10.0 V				$\overline{C}$ to Q_5							
		204													GND	10.0 V	OUT		$\overline{C}$ to Q_6							
t_{PLH1}		205	10.0 V	OUT	IN_2											GND			CLK to Q_1							
		206			GND	IN_2	OUT												CLK to Q_2							
		207				GND		IN_2	OUT										CLK to Q_3							
		208						GND				OUT	IN_2						CLK to Q_4							
		209											GND	OUT	IN_2				CLK to Q_5							
		210													GND	IN_2	OUT		CLK to Q_6							
t_{THL}	3004 Fig. 6	211		OUT	IN_2											GND			Q_1		100					µs
		212			GND	IN_2	OUT									GND			Q_2							
		213			GND	GND		IN_2	OUT							GND			Q_3							

See footnotes at end of device type 05.

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TABLE III. Group A inspection for device type 05 -Continued.

Symbol	MIL-STD-883 method	Cases E,F,Z	Terminal conditions 1/															Measured terminal	Test limits						Units
																			Subgroup 12 T_A = 25°C						
		Symbol	C	Q_1	D_1	D_2	Q_2	D_3	Q_3	V_{SS}	CLK	Q_4	D_4	Q_5	D_5	D_6	Q_6	V_{DD}								
		Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Min	Max	Min	Max	Min	Max	
t_{THL} t_{THL} t_{THL} t_{TLH}	3004 Fig. 6	214 215 216 217 218 219 220 221 222	10.0 V	OUT	GND GND GND IN_2 GND	GND IN_2 GND	OUT	GND IN_2 GND GND GND	OUT	GND	IN_1	OUT OUT	IN_2 GND IN_2 GND GND	OUT OUT	GND IN_2 GND IN_2 GND	GND GND IN_2 GND IN_2	OUT OUT	10.0 V	Q_4 Q_5 Q_6 Q_1 Q_2 Q_3 Q_4 Q_5 Q_6	5	100					us

GS

1/ Pins not designated may be "high" level logic, "low" level logic or open. Exceptions are as follows: $V_{IC(pos)}$ tests, the V_{SS} terminals shall be open; $V_{IC(neg)}$ tests, the V_{DD} terminal shall be open; I_{SS} tests, the output terminals shall be open.

2/ The I_{SS} measurements shall be performed in sequence.

3/ The device manufacturer may, at his option, measure I_{IL} and I_{IH} at 25°C for each individual input or measure all inputs together.

4/ The truth table tests shall be performed in sequence.

5/ The truth table tests shall be performed at V_{IH} and $V_{DD} \leq 5$ Vdc and ≥ 18 Vdc. "L" = V_{SS} + 0.5 V maximum and "H" = V_{DD} -0.5 V minimum.

6/ See 4.4.1c.

7/ Apply clock pulse V_{IN} = 0 to 18 Vdc.

8/ Apply clock pulse V_{IN} = 0 to 15 Vdc.

9/ Apply clock pulse V_{IN} = 0 to 5 Vdc.

10/ The minimum clock frequency (f_{CLK}) requirement is considered met if proper output state changes occur with the pulse repetition period set to that given in the limits column.

11/ The minimum clock, or clear pulse width ($t_{P(CLK)}$, and $t_{P(CLR)}$ requirement, is considered met if proper output state changes occur with the pulse width set to that given in the limits column.

12/ Pulse repetition period = 100 us, 50 percent duty cycle. The maximum clock transition time $t_{TLH(CLK)}$, $t_{THL(CLK)}$ requirement is considered met if proper output state changes occur with rise time set to that given in the limits column.

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d. For class S devices, post dynamic burn-in, or class B devices, post static burn-in, electrical parameter measurements may, at the manufacturer's option, be performed separately or included in the final electrical parameter requirements.

4.2.1 Percent defective allowable (PDA).

a. The PDA for class S devices shall be 5 percent for static burn-in and 5 percent for dynamic burn-in, based on the exact number of devices submitted to each separate burn-in.

b. Static burn-in I and II failures shall be cumulative for determining PDA.

c. The class B devices PDA shall be in accordance with MIL-M-38510 for static burn-in. Dynamic burn-in is not required.

d. Those devices whose measured characteristics after burn-in exceed the specified delta (Δ) limits or electrical parameter limits specified in table III, subgroup 1, are defective and shall be removed from the lot. The verified failures divided by the total number of devices in lot initially submitted to burn-in shall be used to determine the percent defective for the lot and the lot shall be accepted or rejected based on the specified PDA.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, D and E inspections (see 4.4.1 through 4.4.5).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510 and as specified herein. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, D, and E inspections (see 4.4.1 through 4.4.5).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be performed in accordance with table II herein.

b. Subgroups 5 and 6 of table I of method 5005 of MIL-STD-883 shall be omitted.

c. Subgroup 4 (C_i measurement) shall be measured only for initial qualification and after process or design changes which may affect input capacitance. Capacitance shall be measured between the designated terminal and V_{SS} at a frequency of 1 MHz.

d. Subgroup 12 shall be added to the group A inspection requirements for class S devices using an LTPD of 15 and consist of the procedures, test conditions, and limits specified in table III.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883 and as follows:

a. Class S steady state life (accelerated) test circuits shall be submitted to the qualifying activity for approval. When the alternate steady state life test is used, the circuit on figure 5, or equivalent, shall be used.

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b. A special subgroup shall be added using an LTPD of 15 for classes S and B, and shall be measured only for initial qualification and after process or design changes. This subgroup shall consist of a high voltage test of the input protection circuits, V_{ZAP} (see 4.5.3).

c. End-point electrical parameters shall be as specified in table II herein and shall consist only of those subgroups specified in table IIa of test method 5005 of MIL-STD-883, and table II herein also. Delta limits shall apply only to subgroup 5 of group B inspections and shall consist of tests specified in table IV herein.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein. Delta limits shall apply only to subgroup 1 of group C inspection and shall consist of tests specified in table IV herein.

b. Steady-state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition D and as specified in 4.5.2 and as shown on figure 5, or equivalent.

(2) T_A = +125°C minimum.

(3) Test duration, 1,000 hours, except as permitted by appendix B of MIL-M-38510 and method 1005 of MIL-STD-883.

c. Subgroups 3 and 4 shall be added to the group C inspection requirements for class B devices, and shall consist of the tests, conditions, and limits as specified for subgroups 10 and 11 of group A.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

4.4.5 Group E inspection. Group E inspection is required only for device types intended to be marked as radiation hardened (see 3.6.1). When group E testing is performed it shall be in accordance with table V of method 5005 of MIL-STD-883 and 4.5.5 herein.

4.5 Methods of inspection. Methods of inspection shall be specified as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit V_{SS} terminal, unless otherwise specified. Currents given are conventional current and positive when flowing into the referenced terminal.

4.5.2 Burn-in and life test cool down procedures. When these tests are completed and prior to removal of bias voltages, the devices under test (DUT) shall be cooled to a temperature of 25°C ±3°C; then, electrical parameter end-point measurements shall be performed.

TABLE IV. Delta limits at 25°C.

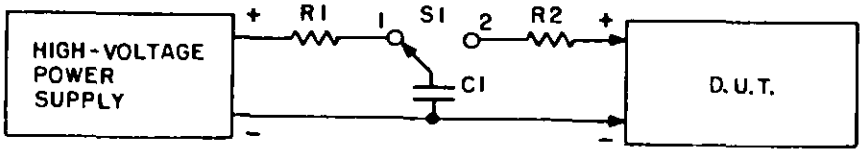
Parameter	Device types				
	01	02	03	04	05
I_{SS}	±10 nA	±10 nA	±10 nA	±10 nA	±10 nA
I_{OL1}	±15%	±15%	±15%	±15%	±15%
I_{OH1}	±15%	±15%	±15%	±15%	±15%

NOTE: Each of the above parameters shall be recorded before and after the required burn-in and life tests to determine delta's (Δ).

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4.5.3 High voltage (V_{ZAP}) test of input protection circuits. All input terminals (up to a maximum of 4) of the DUT shall be subjected to a voltage pulse from a 100 pF source charged to 400 V. This destructive test shall be conducted as follows using the test circuit on figure 7.

a. Measure I_{IL} and I_{IH} at the inputs selected, as stated above, at 25°C. The test limit for each input tested shall be ±10 nA at the specified V_{DD} = 18 V dc. Measure I_{SS} on the DUT at 25°C. The test limit for this measurement shall be increased a maximum of 20 percent of the specified I_{SS} table III limit at the specified V_{DD} = 18 V dc.



V_{ZAP} = 400 V minimum charge on C1
1 MΩ ≤ R1 ≤ 50 M
R2 = 1.5 k
C1 = 100 pF
S1 = Hg-wetted "bounceless" relay

FIGURE 7. High voltage (V_{ZAP}) test circuit.

b. V_{ZAP} is applied to DUT in the following modes (see table V) by changing C1 to V_{ZAP} with S1 in position 1 and then switching to position 2.

TABLE V. Modes for high voltage test.

Mode	+ Terminal	- Terminal
1	V_{DD}	Input
2	Input	V_{SS}
3	Input	Associated Output

c. Within 24 hours repeat the I_{SS}, I_{IL}, and I_{IH} measurements on the same terminals as performed above. If a DUT exhibits leakage currents in excess of the specified limits after the V_{ZAP} test, it shall be classified as a failure.

4.5.4 Quiescent supply current (I_{SS}) test. When performing quiescent supply current measurements (I_{SS}), the meter shall be placed so that all currents flow through the meter.

4.5.5 Radiation hardness assurance (RHA) testing. The RHA testing shall be performed in accordance with test procedures and sampling specified in table V of method 5005 of MIL-STD-883 and herein:

a. Before irradiation, selected samples shall be assembled in qualified packages and pass the governing electrical parameters (group A subgroup 1 at 25°C) and also be subjected to the threshold-voltage test in table VIII in order to calculate the delta threshold (ΔV_T) after irradiation.

b. The devices shall be subjected to a total radiation dose as specified in MIL-M-38510 for the radiation hardness assurance (RHA) level being tested, and meet the end point electrical parameters as defined in table VI at 25°C, after exposure. The start and completion of the end point electrical parameter measurements shall not exceed 2 hours following irradiation.

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c. Threshold-voltage test circuit conditions shall be as specified in table VIII and figure 8. In situ and remote testing, the tests shall be performed with the devices biased in accordance with table VII and bias may be interrupted for up to 1 minute to remove devices to the remote bias fixture.

d. After irradiation, the devices shall pass the truth table test as specified in subgroup 7 in table III or if subgroup 7 is not required, then an equivalent truth table test shall be performed.

TABLE VI. Radiation hardness end-point electrical parameters at 25°C.

Parameter	All device types	V_{DD}
V_{TN}	0.3 V min	10 V
V_{TP}	2.8 V max	10 V
V_T	1.4 V max	10 V
I_{SS}	100 x max limit	18 V
t_{PLH}	1.35 x max limit	5 V
t_{PHL}	1.35 x max limit	5 V

TABLE VII. Bias during exposure to radiation.

Device type	Pin connections		
	10 Vdc through a 30 kΩ to 60 kΩ resistor	V_{SS} = GND	V_{DD} = 10 Vdc
01	1,2,7,9,10,11,12,13,14,15	8	16
02	2,3,4,5,9,10,11,12,13	7	14
03	2,3,4,5,9,10,11,12,13	7	14
04	3,4,5,11,12,13	8	16
05	1,3,4,6,9,11,13,14	8	16

NOTE: Pins not designated are open or may be tied to 10 Vdc through a 30 kΩ to 60 kΩ resistor.

4.6 Data reporting. When specified in the purchase order or contract, a copy of the following data, as applicable, shall be supplied.

a. Attributes data for all screening tests (see 4.2) and variables data for all static burn-in, dynamic burn-in, and steady state life tests (see3.5).

b. A copy of each radiograph.

c. The quality conformance inspection data (see 4.4).

d. Parameter distribution data on parameters evaluated during burn-in (see 3.5).

e. Final electrical parameters data (see 4.2c).

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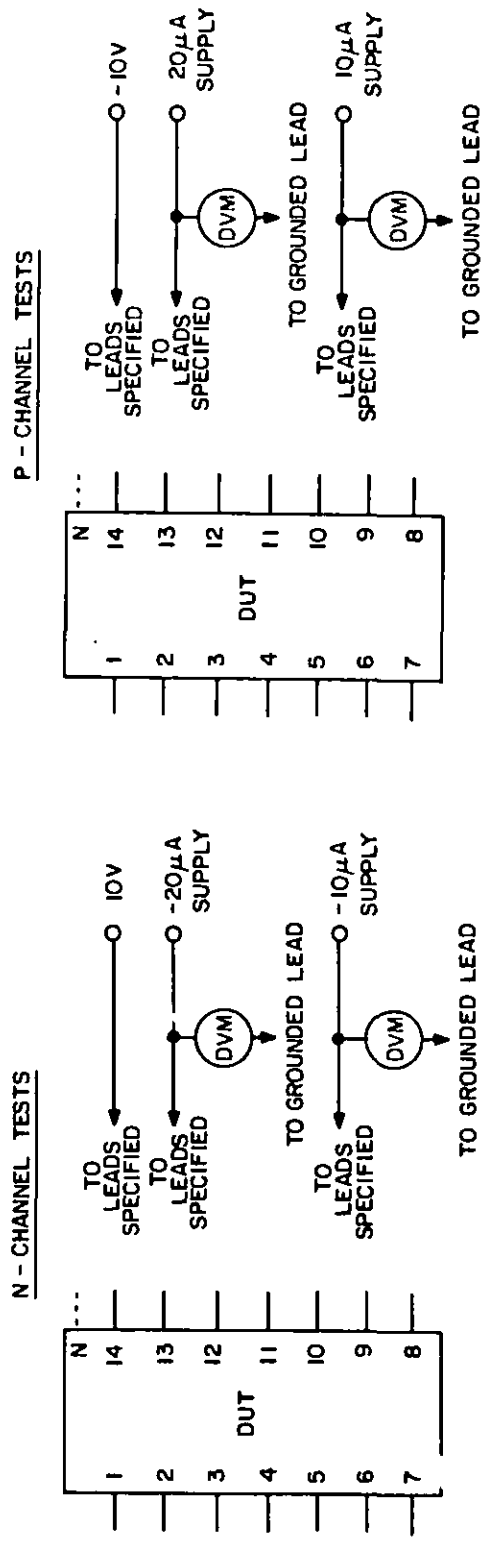


FIGURE 8. Threshold-voltage test circuit.

TABLE VIII. Threshold - voltage test circuit conditions.

Device type	GND	10V	V_{TN} measured at		GND	-10V	V_{TP} measured at	
			-20 μA supply	-10 μA supply			20 μA supply	10 μA supply
01	7	1,2, 9-16		8	7	1, 2, 8-15		16
02	12	14		2-5, 7, 9-11,13	12	2-5,7, 9-11,13		14
03	12	14		2-5, 7, 9-11,13	12	2-5,7, 9-11,13		14
04	4	5, 11, 12, 16		1, 3, 8, 13,15	4	1, 3, 8, 13,15		5,11,12,16
05	3	1,4,6,9,11,13, 14,16		8	3	8		1,4,6,9,11, 13,14,16

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design application and logisitic support of existing equipment.

6.2 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action and reporting of results, if applicable.

f. Requirements for quality assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements shall not apply to direct purchase by, or direct shipment to the Government.

h. Requirements for "JAN" marking.

i. Requirements for total dose radiation testing (see 3.6.1 and 4.5.5), if applicable.

6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

C_i	Input terminal-to-V_{SS} capacitance.
GND	Ground zero voltage potential.
I_{OC}	3-state output leakage current.
I_{SS}	Quiescent supply current.
T_A	Free air temperature.
V_{DD}	Positive supply voltage.
V_{SS}	Negative supply voltage.
V_{ZAP}	Input test voltage.

6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class S for National Aeronautics and Space Administration or class B for Department of Defense (see 1.2.2), lead finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

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Military device type	Generic-industry type
01	4076B
02	4095B
03	4096B
04	4098B
05	40174B

6.6 Handling. MOS devices must be handled with certain precautions to avoid damage due to accumulation of static charge. Input protective devices have been designed in the chip to minimize the effect of this static build up. However, the following handling practices are recommended:

a. Devices should be handled on benches with conductive and grounded surface.

b. Ground test equipment and tools.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, or silk in MOS areas.

f. Maintain relative humidity above 50 percent, if practical.

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - MI
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - AR, SM
Navy - AS, CG, MC, OC, SH

Agent:
DLA - ES

Preparing activity:
NASA - NA

(Project 5962-0639-6)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-38510/175B	MICROCIRCUITS, DIGITAL, CMOS, POSITIVE LOGIC, FLIP-FLOPS AND MONOSTABLE MULTIVIBRATOR, MONOLITHIC SILICON

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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